Answer Key

Other (162 Questions)

Q1. (D)	Q2. (B)	Q3. (A)	Q4. (B)	Q5. (B)
Q6. (A)	Q7. (B)	Q8. (A)	Q9. (B)	Q10. (A)
Q11. (A)	Q12. (C)	Q13. (C)	Q14. (B)	Q15. (A)
Q16. (D)	Q17. (A)	Q18. (A)	Q19. (D)	Q20. (C)
Q21. (B)	Q22. (D)	Q23. (B)	Q24. (D)	Q25. (D)
Q26. (B)	Q27. (C)	Q28. (A)	Q29. (A)	Q30. (B)
Q31. (B)	Q32. (D)	Q33. (D)	Q34. (D)	Q35. (B)
Q36. (C)	Q37. (C)	Q38. (A)	Q39. (A)	Q40. (C)
Q41. (D)	Q42. (C)	Q43. (D)	Q44. (A)	Q45. (B)
Q46. (A)	Q47. (A)	Q48. (C)	Q49. (D)	Q50. (B)
Q51. (C)	Q52. (B)	Q53. (D)	Q54. (C)	Q55. (B)
Q56. (D)	Q57. (D)	Q58. (D)	Q59. (D)	Q60. (C)
Q61. (A)	Q62. (C)	Q63. (A)	Q64. (A)	Q65. (B)
Q66. (A)	Q67. (C)	Q68. (D)	Q69. (C)	Q70. (B)
Q71. (C)	Q72. (A)	Q73. (B)	Q74. (B)	Q75. (B)
Q76. (B)	Q77. (C)	Q78. (D)	Q79. (A)	Q80. (B)
Q81. (D)	Q82. (C)	Q83. (D)	Q84. (B)	Q85. (B)
Q86. (A)	Q87. (D)	Q88. (C)	Q89. (C)	Q90. (D)
Q91. (C)	Q92. (A)	Q93. (B)	Q94. (C)	Q95. (C)
Q96. (D)	Q97. (D)	Q98. (A)	Q99. (A)	Q100.(D)
Q101.(B)	Q102.(C)	Q103.(D)	Q104.(B)	Q105.(D)

Q106.(C)	**Q107.(C)**	**Q108.(A)**	**Q109.(B)**	**Q110.(D)**
Q111.(C)	**Q112.(A)**	**Q113.(C)**	**Q114.(A)**	**Q115.(C)**
Q116.(B)	**Q117.(C)**	**Q118.(A)**	**Q119.(B)**	**Q120.(A)**
Q121.(D)	**Q122.(C)**	**Q123.(D)**	**Q124.(C)**	**Q125.(C)**
Q126.(B)	**Q127.(D)**	**Q128.(A)**	**Q129.(A)**	**Q130.(A)**
Q131.(A)	**Q132.(A)**	**Q133.(B)**	**Q134.(A)**	**Q135.(D)**
Q136.(B)	**Q137.(D)**	**Q138.(B)**	**Q139.(C)**	**Q140.(B)**
Q141.(C)	**Q142.(D)**	**Q143.(A)**	**Q144.(A)**	**Q145.(A)**
Q146.(B)	**Q147.(A)**	**Q148.(A)**	**Q149.(B)**	**Q150.(D)**
Q151.(C)	**Q152.(B)**	**Q153.(A)**	**Q154.(C)**	**Q155.(C)**
Q156.(B)	**Q157.(C)**	**Q158.(D)**	**Q159.(C)**	**Q160.(A)**
Q161.(C)	**Q162.(B)**			

Solutions

Q1. Solution

Correct Answer: (D)

$\mathrm{V}=5x^2+10x-4$
$\mathrm{E}=\frac{-\mathrm{dV}}{\mathrm{dx}}=-(10x+10)$
At $x=1m, \mathrm{E}=-20\ \mathrm{V}/m$

Q2. Solution

Correct Answer: (B)

$r=\frac{mv\sin\theta}{\mathrm{B}e}=\frac{3\times10^5\sin30^\circ}{0.3\times10^8}$
$\frac{3\times10^5\times\frac{1}{2}}{3\times10^7}=0.5\times10^{-2}$
$m=0.5\ \mathrm{cm}$

Q3. Solution

Correct Answer: (A)

Let $\mathrm{m}=\mathrm{KF^aL^bT^c}$
Substituting the dimensions of
$[\mathrm{F}]=\left[\mathrm{MLT}^{-2}\right],[\mathrm{L}]=[\mathrm{L}]$ and $[\mathrm{T}]=[\mathrm{T}]$ and comparing both side, we get $\mathrm{m}=\mathrm{FL}^{-1}\ \mathrm{T}^2$

Q4. Solution

Correct Answer: (B)

Induced emf produced between the centre and a point on the disc is given by
$e=\frac{1}{2}\omega\mathrm{BR}^2$
$\omega=60\mathrm{rad/s}, \mathrm{B}=0.05\ \mathrm{Wb/m^2}$
and $\mathrm{R}=100\ \mathrm{cm}=1\ \mathrm{m}$
We get $e=\frac{1}{2}\times60\times0.05\times(1)^2=1.5\ \mathrm{V}$

Q5. Solution

Correct Answer: (B)

For stone to be dropped from rising balloon of velocity $29\ \mathrm{m/s}, \mathrm{u}=-29\ \mathrm{m/s}, \mathrm{t}=10\mathrm{sec}$.
$\therefore \mathrm{h}=-29\times10+\frac{1}{2}\times9.8\times100$
$=-290+490=200\ \mathrm{m}$.

Q6. Solution

Correct Answer: (A)

The phase angle between voltage V and current I is $\frac{\pi}{2}$.
Therefore, power factor $\cos\phi = \cos\left(\frac{\pi}{2}\right) = 0$.
Hence the power consumed is zero.

Q7. Solution

Correct Answer: (B)

The bullets are fired at the same initial speed
$\frac{\mathrm{H}}{\mathrm{H}'} = \frac{\mathrm{u}^2 \sin^2 60^\circ}{2\,\mathrm{g}} \times \frac{2\,\mathrm{g}}{\mathrm{u}^2 \sin^2 30^\circ} = \frac{\sin^2 60^\circ}{\sin^2 30^\circ}$
$= \frac{(\sqrt{3}/2)^2}{(1/2)^2} = \frac{3}{1}$

Q8. Solution

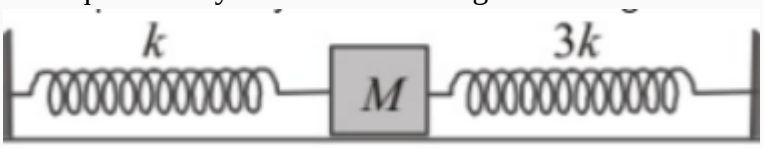
Correct Answer: (A)

$\mathrm{a}^{\lambda}\mathrm{g} = \frac{\lambda_{\mathrm{a}}}{\mu} = \frac{5460}{1.5} = 3640\mathring{A}$

Q9. Solution

Correct Answer: (B)

The equivalent system is shown in figure.



The equivalent force constant of which $k_{\mathrm{e}} = k + 3k = 4k \therefore \omega = \sqrt{\frac{k_e}{m}} = \sqrt{\frac{4k}{m}} = 2\sqrt{\frac{k}{m}}$
and $\mathrm{f} = \frac{\omega}{2\pi} = \frac{2}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{\pi}\sqrt{\frac{k}{m}}$

Q10. Solution

Correct Answer: (A)

According to kepler's law of area $\frac{\mathrm{dA}}{\mathrm{dt}} = \frac{\mathrm{L}}{2\,\mathrm{m}}$
For central forces, torque $= 0$
$\therefore \mathrm{L} = \text{constant}$
$\therefore \frac{\mathrm{dA}}{\mathrm{dt}} = \text{constant}$

Q11. Solution

Correct Answer: (A)

If N, is the total number of particles in the sample and $\lambda = \log_e 2/T$ is the decay constant.The activity-Ac, is the number of decays per unit time of a radioactive sample. It is given as $\mathrm{Ac} = \lambda \mathrm{N} \Rightarrow \mathrm{N} = \mathrm{Ac}/\lambda -$ (1) As material is same for all the specimens, the decay constant λ is same for all. Now, one curie is 1 Curie $= 3.7 \times 10^{10}$ decays $/$sec 1 rutherford $= 2.7 \times 10^{-5}$ Curie 1 Becquerel $= 2.7 \times 10^{-11}$ Curie So the specimen A has the maximum activity, and from (1), the number of particles or total mass is directly proportional to the activity. So specimen A has the maximum mass. ~

Q12. Solution

Correct Answer: (C)

Length of solenoid, $l = 2$ m
Total number of turns, $\mathrm{N} = 5 \times 1000 = 5000$
Number of turns per unit length,
$\mathrm{n} = \frac{\mathrm{N}}{l} = \frac{5000}{2} = 2500\ \mathrm{m}^{-1}$
Current, I = 5A
Now, $\mathrm{B} = \mu_0 \mathrm{nI} = 4\pi \times 10^{-7} \times 2500 \times 5\ \mathrm{T}$
$= 1.57 \times 10^{-2}\ \mathrm{T}$,

Q13. Solution

Correct Answer: (C)

The tension in the above string can be divided into two components where the component opposite to weight will balance the weight while the component of this tension opposite to T_1 will be equal to T_1.
$\mathrm{T} \sin\theta = 2$ kgwt
$\mathrm{T} \cos\theta = \mathrm{T}_1$
The first equation gives
$\mathrm{T} = 4$ kgwt
So T_1 will be,
$\mathrm{T}_1 = 2\sqrt{3}$ kgwt ~

Q14. Solution

Correct Answer: (B)

$\mathrm{R} = \frac{\rho \mathrm{l}}{\mathrm{A}}$
Now, $1 = 1 + \frac{1}{10} = \frac{111}{10}$
and therefore, $\mathrm{A} = \frac{10\ \mathrm{A}}{11}$
So $\mathrm{R}' = \frac{\mathrm{p} \times \left(\frac{111}{10}\right)}{\frac{10\ \mathrm{A}}{11}} = \frac{\rho \mathrm{l}}{\mathrm{A}} \times \frac{(11)^2}{(10)^2} = 1.21\mathrm{R}$
Now resistance becomes 1.21 times of initial and specific resistance is the intrinsic property so remains same. ,

Q15. Solution

Correct Answer: (A)

The significant result deduced from the Rutherford's scatter ing is that whole of the positive charge is concentrated at the centre of atom i.e. nucleus. .

Q16. Solution

Correct Answer: (D)

Let the velocity of the ball just when it leaves the hand is u then applying,
$v^2 - u^2 = 2a$ for upward journey
$\Rightarrow -u^2 = 2(-10) \times 2$
$\Rightarrow u^2 = 40$
Again applying $v^2 - u^2 = 2as$ for the upward journey of the ball, when the ball is in the hands of the thrower,
$v^2 - u^2 = 2as$
$\Rightarrow 40 - 0 = 2(a)0.2$
$\Rightarrow a = 100 \text{ m/s}^2$
$\therefore \mathrm{F} = ma = 0.2 \times 100 = 20 \text{ N}$ ^
$\Rightarrow \mathrm{N} - mg = 20$
$\Rightarrow \mathrm{N} = 20 + 2 = 22 \text{ N}$

Q17. Solution

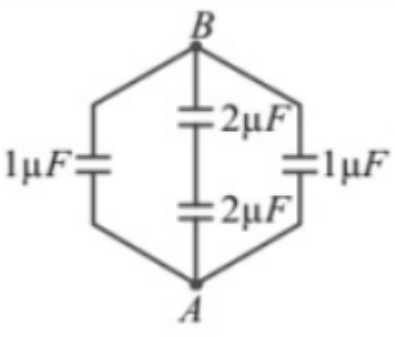
Correct Answer: (A)

Here, $E = 9 \text{ V}; \mathrm{V_z} = 6; \mathrm{R_L} = 1000\Omega$ and $\mathrm{R_s} = 100\Omega$,
Potential drop across series resistor $\mathrm{V} = \mathrm{E} - \mathrm{V_Z} = 9 - 6 = 3 \text{ V}$
Current through series resistance R_S is $\mathrm{I} = \frac{\mathrm{V}}{\mathrm{R}} = \frac{3}{100} = 0.03 \text{ A}$
Current through load resistance R_L is $\mathrm{I_L} = \frac{\mathrm{V_Z}}{\mathrm{R_L}} = \frac{6}{1000} = 0.006 \text{ A}$
Current through Zener diode is $\mathrm{I_Z} = \mathrm{I} - \mathrm{I_L} = 0.03 - 0.006 = 0.024\text{amp}$. Power dissipated in Zener diode is $\mathrm{P_Z} = \mathrm{V_Z I_Z} = 6 \times 0.024 = 0.144\text{Watt}$ ^

Q18. Solution

Correct Answer: (A)

The equivalent circuit is shown in figure.



$\mathrm{C_{AB}} = 1 + \frac{2\times2}{2+2} + 1$
$\mathrm{C_{AB}} = 3\mu\mathrm{F}$ ^

Q19. Solution

Correct Answer: (D)

If l is length of pendulum and θ be angular amplitude then from principle of conservation of energy (Initial total energy) $0 + \mathrm{mgh} = \frac{1}{2}mv^2 + 0$ (Final total energy) $\Rightarrow v = \sqrt{2gh} = \sqrt{2gl(1-\cos\theta)}$!

Q20. Solution

Correct Answer: (C)

$\mathrm{R} = \frac{\mathrm{V}}{\mathrm{I_g}} - \mathrm{G} = \frac{10}{0.01} - 10 = 990\Omega$ in series. ,

Q21. Solution

Correct Answer: (B)

$n' = \frac{nv}{v-v_s} \ldots\ldots\ldots (1)$

$n'' = \frac{nv}{v+v_s} \ldots\ldots\ldots (2)$

From (1) and (2), $\frac{n'}{n''} = \frac{v+v_s}{v-v_s} \ldots\ldots\ldots (3)$

According to question, $\frac{n'}{n'} = \frac{5}{4}$

$v_s = ?v = 330 \text{ m/s} \ldots\ldots\ldots\ldots (4)$

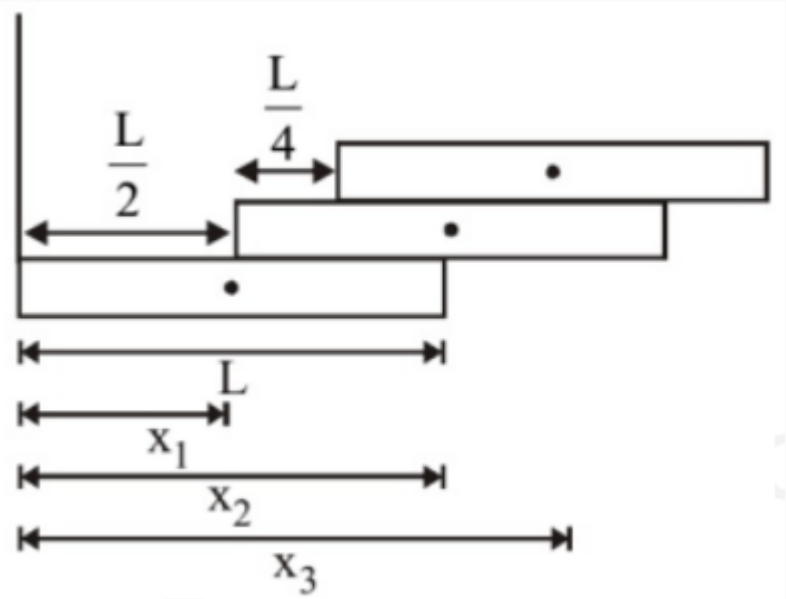
From eq. (3) and (4)

$\frac{5}{4} = \left[\frac{330+v_s}{330-v_s}\right] \Rightarrow 9v_s = 330$

$\therefore v_s = 36.6 \text{ m/s}$

Q22. Solution

Correct Answer: (D)



$x_1 = \frac{\mathrm{L}}{2}, x_2 = \mathrm{L}, x_3 = \frac{5\,\mathrm{L}}{4}$

$\therefore x_{\mathrm{CM}} = \frac{m_1x_1+m_2x_2+m_3x_3}{m_1+m_2+m_3}$

$= \frac{\mathrm{M}\times\frac{\mathrm{L}}{2}+\mathrm{M}\times\mathrm{L}+\mathrm{M}\times\frac{5\,\mathrm{L}}{4}}{\mathrm{M}+\mathrm{M}+\mathrm{M}} = \frac{11\,\mathrm{L}}{12}$

Q23. Solution

Correct Answer: (B)

The figure is showing I - V characteristics of non ohmic or non linear conductors.

Q24. Solution

Correct Answer: (D)

$\tan\theta = \frac{\mathrm{V}}{\mathrm{H}} = \frac{3}{4}\left[\because \tan 37^\circ = \frac{3}{4}\right]$
$\therefore \mathrm{V} = \frac{3}{4}\mathrm{H}$
$\mathrm{V} = 6 \times 10^{-5}\ \mathrm{T}$
$\mathrm{H} = \frac{4}{3} \times 6 \times 10^{-5}\ \mathrm{T} = 8 \times 10^{-5}\ \mathrm{T}$
$\therefore \mathrm{B}_{\text{total}} = \sqrt{\mathrm{V}^2 + \mathrm{H}^2} = \sqrt{(36+64)} \times 10^{-5}$
$= 10 \times 10^{-5} = 10^{-4}\ \mathrm{T}.$

Q25. Solution

Correct Answer: (D)

Total angular momentum of the system initially $\mathrm{L_i} = \mathrm{I_1 w} + \mathrm{I_2}(0) = \mathrm{I_1 w}$
Total angular momentum of the system finally $L_f = (I_1 + I_2)w_2$
According to conservation of angular momentum i.e. $\mathrm{L_i} = \mathrm{L_f}$
$\Rightarrow \mathrm{I_1}\omega = (\mathrm{I_1} + \mathrm{I_2})\omega_2$
$\Rightarrow \omega_2 = \dfrac{\mathrm{I_1}\omega}{\mathrm{I_1} + \mathrm{I_2}}$

Q26. Solution

Correct Answer: (B)

Sharpness of resonance of a resonant L-C-R circuit is determined by the ratio of resonant frequency with the selectivity of the circuit. This ratio is also called "Quality Factor" or Q-factor.
$Q\text{-factor} = \frac{\omega_0}{2\Delta\omega} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$

Q27. Solution

Correct Answer: (C)

$\frac{\mathrm{dQ}}{\mathrm{dt}} = \frac{\mathrm{kA}(\mathrm{T_1} - \mathrm{T_2})}{\mathrm{L}}$ $[(\mathrm{T_1} - \mathrm{T_2})$ is the temperature difference $]$

Q28. Solution

Correct Answer: (A)

Weight of liquid of height $\mathrm{H}=\frac{\pi \mathrm{d}^2}{4} \times \mathrm{H} \rho \times \mathrm{g} \rightarrow$ (i)

Let us consider a mass dm situated at a distance x from A as shown in the figure. The centripetal force required for the mass to rotate $=(\mathrm{dm}) \omega^2$

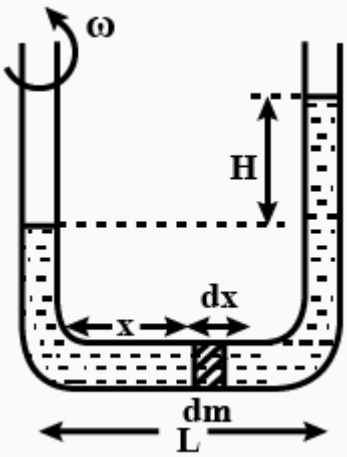
$\therefore$ The total centripetal force required for the mass of length L to rotate

$=\int_0^1\left(\rho \mathrm{x} \frac{\pi \mathrm{d}^2}{4}\right) \mathrm{xdx} \times\left(\mathrm{x} \omega^2\right)=\rho \times \frac{\pi \mathrm{d}^2}{4} \times \omega^2 \times \frac{1^2}{2} \rightarrow$ (ii)

This centripetal force required is provided by the weight of liquid of height 'H' From (i) and (ii)

$$\frac{\pi \mathrm{d}^2}{4} \times \mathrm{H} \times \rho \times \mathrm{g}=\rho \times \frac{\pi \mathrm{d}^2}{4} \times \omega^2 \times \frac{1^2}{2}$$

$$\Rightarrow \mathrm{H}=\frac{\omega^2 \mathrm{l}^2}{2 \mathrm{~g}}$$



Q29. Solution

Correct Answer: (A)

$\lambda=667 \times 10^{-9} m, \mathrm{P}=9 \times 10^{-3}$ W

$\mathrm{P}=\frac{\mathrm{Nhc}}{\lambda}, \mathrm{N}=$ No. of photons emitted/sec.

$\mathrm{N}=\frac{9 \times 10^{-3} \times 667 \times 10^{-9}}{6.6 \times 10^{-34} \times 3 \times 10^8}=3 \times 10^{16} / \mathrm{sec}$

Q30. Solution

Correct Answer: (B)

From equation given,

$\omega=100$

and $\mathrm{k}=20$

$\mathrm{v}=\frac{\omega}{\mathrm{k}}=\frac{100}{20}=5 \mathrm{~m} / \mathrm{s}$

Q31. Solution

Correct Answer: (B)

Volume $v = \frac{\mathrm{m}}{d}$ and using $\mathrm{PV}^{\gamma} =$ constant

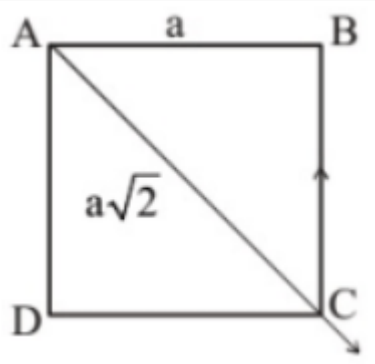
$\frac{\mathrm{P}'}{\mathrm{P}} = \frac{\mathrm{V}}{\mathrm{V}'} = \left(\frac{\mathrm{d}'}{\mathrm{d}}\right)^{\gamma}$

or $128 = (32)^{\gamma}$

$\therefore \gamma = \frac{7}{5} = 1.4$

Q32. Solution

Correct Answer: (D)



It is given that $\vec{F}_A + \vec{F}_B + \vec{F}_D = 0$

Where $\vec{F}_A, \vec{F}_B$ and $\vec{F}_D$ are the forces applied by charges placed at A, B and D on the charge placed at C.

$\Rightarrow \overrightarrow{\mathrm{F}}_{\mathrm{B}} + \overrightarrow{\mathrm{F}}_{\mathrm{D}} = -\overrightarrow{\mathrm{F}}_{\mathrm{A}}$

$\overrightarrow{\mathrm{F}}_{\mathrm{B}} + \overrightarrow{\mathrm{F}}_{\mathrm{D}} = \sqrt{2}\frac{\mathrm{KqQ}}{\mathrm{a}^2}$

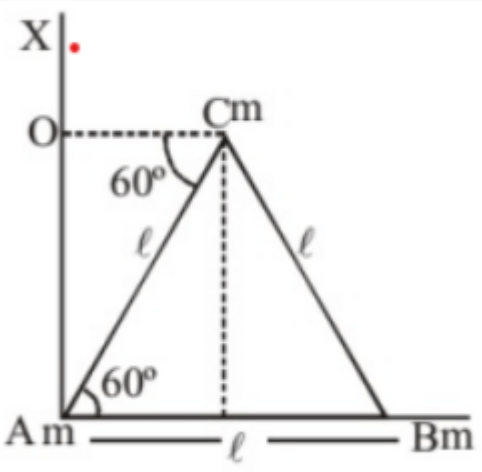
$\Rightarrow \overrightarrow{\mathrm{F}}_{\mathrm{A}} = \frac{\mathrm{KQ}^2}{2a^2}$

$\therefore \sqrt{2}\frac{\mathrm{KqQ}}{a^2} = -\frac{\mathrm{KQ}^2}{2a^2}$

$\Rightarrow \frac{\mathrm{Q}}{q} = -2\sqrt{2}$

Q33. Solution

Correct Answer: (D)



$$\mathrm{I_{AX}} = \mathrm{m(AB)}^2 + \mathrm{m(OC)}^2 = \mathrm{m}l^2 + \mathrm{m}(l\cos 60^\circ)^2 = ml^2 + \frac{\mathrm{m}l^2}{4} = \frac{5}{4}\,\mathrm{m}l^2$$

Q34. Solution

Correct Answer: (D)

Energy $=$ charge $\times$ potential diff.

$\mathrm{E_{electron}} = \mathrm{q_e V}$ and $\mathrm{E_{proton}} = \mathrm{q_p} 4\ \mathrm{V}$

de $-$ Broglie wavelength $\lambda = \frac{\mathrm{h}}{\mathrm{P}} = \frac{\mathrm{h}}{\sqrt{2\mathrm{mE}}}$

$\lambda_e = \frac{h}{\sqrt{2m_e \mathrm{eV}}}$ and $\lambda_\mathrm{P} = \frac{h}{\sqrt{2m_\mathrm{p} e 4\ \mathrm{V}}}$ $\because (q_e = q_p)$

$\therefore \frac{\lambda_e}{\lambda_\mathrm{P}} = \frac{\frac{h}{\sqrt{2m_e \mathrm{eV}}}}{\frac{h}{\sqrt{2m_\mathrm{p} e 4\ \mathrm{V}}}} = 2\sqrt{\frac{m_\mathrm{P}}{m_e}}$

Q35. Solution

Correct Answer: (B)

$$\frac{\mathrm{v_1}}{\mathrm{v_2}} = \frac{\sqrt{\frac{3\mathrm{RT_1}}{\mathrm{M}}}}{\sqrt{\frac{3\mathrm{RT_2}}{\mathrm{M}}}} = \sqrt{\frac{\mathrm{T_1}}{\mathrm{T_2}}} \therefore \mathrm{v_2} = 2\mathrm{v_1}$$

Q36. Solution

Correct Answer: (C)

Let ' M ' be the mass of the particle

Now, $E_{\text{initial}} = E_{\text{final}}$

i.e. $\frac{\mathrm{GMm}}{\sqrt{2}r} + 0 = \frac{\mathrm{GM}m}{r} + \frac{1}{2}\mathrm{MV}^2$

or, $\frac{1}{2}\mathrm{MV}^2 = \frac{\mathrm{GM}m}{r}\left[1 - \frac{1}{\sqrt{2}}\right]$

$\Rightarrow \frac{1}{2}\ \mathrm{V}^2 = \frac{\mathrm{G}m}{r}\left[1 - \frac{1}{\sqrt{2}}\right]$ or,

$\mathrm{V} = \sqrt{\frac{2\mathrm{Gm}}{r}\left(1 - \frac{1}{\sqrt{2}}\right)}$

Q37. Solution

Correct Answer: (C)

Incident momentum, $p = \frac{E}{c}$

For perfectly reflecting surface with normal incidence

$\Delta p = 2p = \frac{2\mathrm{E}}{c}; \mathrm{F} = \frac{\Delta p}{\Delta t} = \frac{2\mathrm{E}}{ct}$

Q38. Solution

Correct Answer: (A)

Compressibility $= \frac{1}{\mathrm{B}} = \frac{\Delta \mathrm{V}/\mathrm{V}}{\mathrm{P}}$.

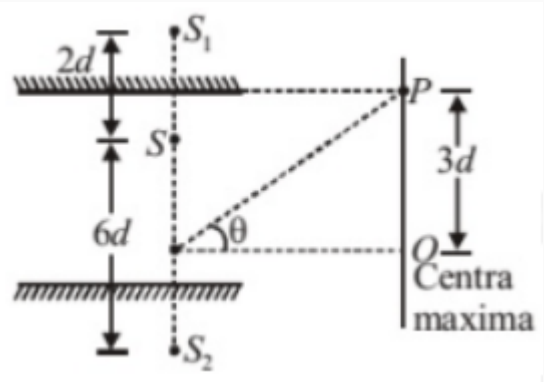
Here, $\mathrm{P} = 100$ atm

Compressibility $= 4 \times 10^{-5}$ and $\mathrm{V} = 100\ \mathrm{cm}^3$

Hence, $\Delta \mathrm{V} = 0.4\ \mathrm{cm}^3$

Q39. Solution

Correct Answer: (A)



At P, $\Delta x = \frac{(8d) \times 3d}{\mathrm{D}}$

For 2 nd maxima, $\Delta x = 2\lambda$

$\Rightarrow \frac{24d^2}{\mathrm{D}} = 2\lambda$

$\Rightarrow \lambda = \frac{12d^2}{\mathrm{D}}$

Q40. Solution

Correct Answer: (C)

Volume of 8 small droplets = Volume of 1 big drop $\Rightarrow \frac{4}{3}\pi r^3 \times 8 = \frac{4}{3}\pi \mathrm{R}^3 \Rightarrow r = \frac{\mathrm{R}}{(8)^{1/3}}$

Work done = (Change in area)$\times$ surface tension $= \left(4\pi r^2 n - 4\pi \mathrm{R}^2\right)\mathrm{T}$

$\left\{4\pi\left(\frac{\mathrm{R}^2}{(8)^{2/3}}\right) \times 8 - 4\pi \mathrm{R}^2\right\}\mathrm{T} = 4\pi \mathrm{R}^2\ \mathrm{T}$

Q41. Solution

Correct Answer: (D)

Fe^{3+} (d^5) has 5 unpaired electrons therefore magneticmoment $= \sqrt{n(n+2)} = \sqrt{5(5+2)} = 5.91$ which is maximum among given options. As $Sc^{3+}, Ti^{3+}, Cr^{3+}$ and V^{3+} contain $0, 1, 3,$ and 2 number of unpaired electrons respectively.

Q42. Solution

Correct Answer: (C)

Bond length $\propto \frac{1}{\text{Bond order}}$ Bond order of $ClO^- = 1$, Bond order of $ClO_2^- = 1.5$, Bond order of $ClO_3^- = 1.66$, Bond order of $ClO_4^- = 1.75$. Thus the order of bond length is
$ClO_4^- < ClO_3^- < ClO_2^- < ClO^-$

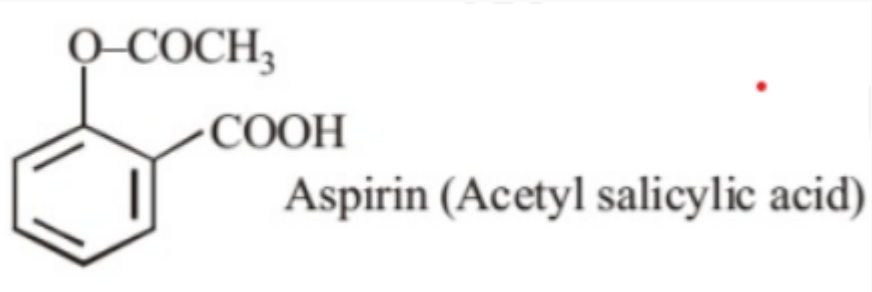
Q43. Solution

Correct Answer: (D)

Cisplatin $[Pt(NH_3)_2Cl_2]$ is used in cancer treatment.

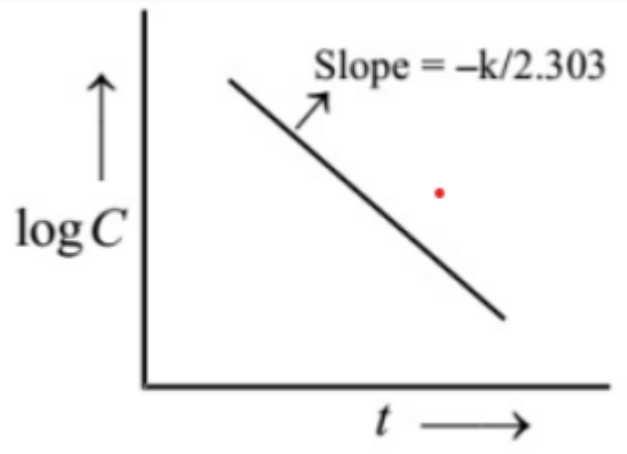
Q44. Solution

Correct Answer: (A)



Q45. Solution

Correct Answer: (B)



Q46. Solution

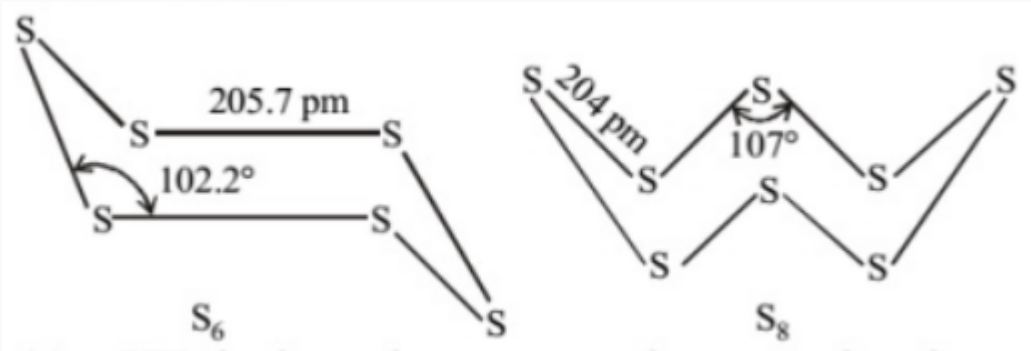
Correct Answer: (A)

Nitrogen due to small size is able to show $p\pi - p\pi$ lateral overlap forming $\mathrm{N} \equiv \mathrm{N}$, rest elements due to bigger size are not able to show $p\pi - p\pi$ lateral overlap.

Q47. Solution

Correct Answer: (A)

The S_6 molecule has a chair-form hexagon ring with the approx same bond length as that in S_8, but with some what smaller bond angles i.e. bond lengths are approx same but bond angles are different.



Q48. Solution

Correct Answer: (C)

HF_2 is the only compound among the given options which does not contain any coordinate bond because it has hydrogen bonding.

$[\mathrm{H} - \mathrm{F} \bullet \bullet \bullet \mathrm{H}]^-$

Q49. Solution

Correct Answer: (D)

$\mathrm{BF}_3, \mathrm{BCl}_3, \mathrm{BBr}_3$ are sp^2 hybridised. So, all have same structure and bond angle i.e. 120°.

Q50. Solution

Correct Answer: (B)

$\mathrm{P}_{\mathrm{A}}^{\circ} = 7 \times 10^3$

$\mathrm{P}_{\mathrm{B}}^{\circ} = 12 \times 10^3$

$x'_{\mathrm{A}} = 0.4; x'_{\mathrm{B}} = 1 - 0.4$

$x'_{\mathrm{B}} = 0.6$

$\mathrm{P}_{\text{total}} = \mathrm{P}_{\mathrm{A}^\circ}^{x'_{\mathrm{A}}} + \mathrm{P}_{\mathrm{B}}^{\circ} x'_{\mathrm{B}}$

$= 7 \times 10^3 \times 0.4 + 12 \times 10^3 \times 0.6$

$= (7 \times 0.4 + 12 \times 0.6) \times 10^3 = 10^4$

$x_{\mathrm{A}} = \frac{\mathrm{P}_{\mathrm{A}^\circ}^{x'}\mathrm{A}}{\mathrm{P}_{\text{total}}} = \frac{7 \times 10^3 \times 0.4}{10^4}$

$\therefore x_{\mathrm{A}} = 0.28, x_{\mathrm{B}} = 1 - 0.28 = 0.72$

Q51. Solution

Correct Answer: (C)

Completing the reaction, we get

$\mathrm{NCl_3 + 3H_2O \rightarrow 3\underset{'x'}{HClO} + NH_3}$

Q52. Solution

Correct Answer: (B)

No. of lattice points = No. of octahedral voids = $\frac{1}{2}\times$ No. of tetrahedral voids in *ccp* structure

$\therefore$ No. of atoms of $B = 4$

No. of atoms of $A = \frac{1}{2}\times$ No. of octahedral voids

$= \frac{1}{2} \times 4 = 2$

No. of atoms of O = All tetrahedral voids = $2\times$ No. of lattice points

$= 2 \times 4 = 8$

Hence, $A : B : O = 1 : 2 : 4$

Therefore, the formula of the compound is AB_2O_4

Q53. Solution

Correct Answer: (D)

Complex X is $[\mathrm{Fe(en)_3}]^{2+}$; as 'en' is a strong field ligand pairing of electrons will take place. $[\mathrm{Fe(en)_3}]^{2+}$: d^2sp^3 hybridisation

Hence, hybridisation is d^2sp^3 and complex is diamagnetic. As it has 3 bidentate symmetrical 'en' ligands so it will not show geometrical isomerism.

Q54. Solution

Correct Answer: (C)

$$\mathrm{v} = \frac{1}{\mathrm{h}} \times \mathrm{IE} \times \left[\frac{1}{{\mathrm{n_1}}^2} - \frac{1}{{\mathrm{n_2}}^2}\right]$$
$$= \frac{2.18 \times 10^{-18}}{6.625 \times 10^{-34}} \times \left[\frac{1}{1} - \frac{1}{16}\right] = 3.08 \times 10^{15}\ \mathrm{s}^{-1}$$

Q55. Solution

Correct Answer: (B)

Cetyltrimethyl ammonium bromide which is a germicide, is a popular cationic detergent.

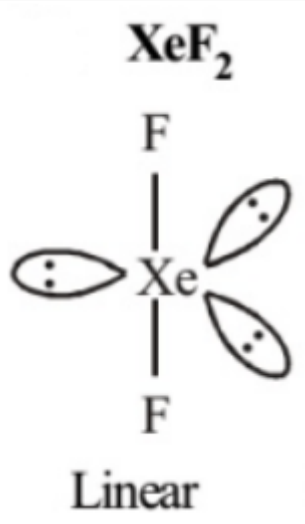
Q56. Solution

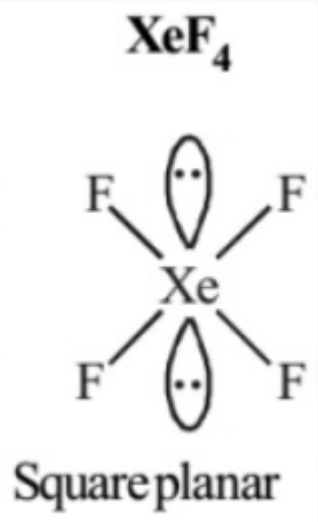
Correct Answer: (D)

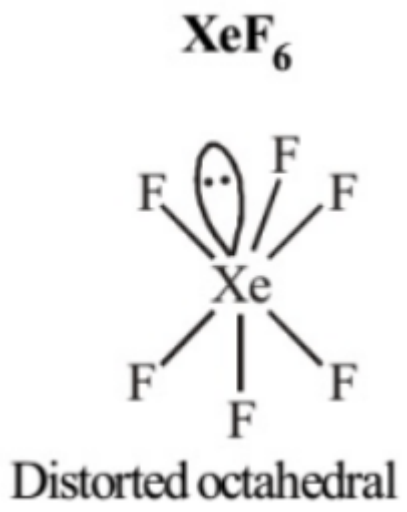
Sucralose is trichloro derivative of sucrose. It is stable at cooking temperture. It does not provide calories.

Q57. Solution

Correct Answer: (D)







Q58. Solution

Correct Answer: (D)

HF form linear polymeric structure due to hydrogen bonding.

Q59. Solution
Correct Answer: (D)

Salvarsan (arsphenamine) is an arsenic-based medicine used for the treatment of syphilis.

Q60. Solution
Correct Answer: (C)

Metel ion	No. of unpaired elec.		Difference in the unpaired electrons
	High spin	Low spin	
$Ni^{2+}(3d^8)$	2	2	0
$Mn^{2+}(3d^5)$	5	1	4
$Fe(3d^6)$	4	0	4
$Co^{2+}(3d^7)$	3	1	2

Q61. Solution
Correct Answer: (A)

According to Freundlich adsorption isotherm $\frac{x}{m} \propto p^{\frac{1}{n}}, \frac{x}{m} = kp^{\frac{1}{n}}$
Slope $= \frac{2}{3}$
$\log \frac{x}{m} = \log k + \frac{1}{n} \log p$
Slope $= \frac{1}{n} = \frac{2}{3}$
$\frac{x}{m} \propto p^{\frac{2}{3}}$

Q62. Solution
Correct Answer: (C)

x has a conjugated diene system, w an isolated diene system, z a cumulated diene system, and y an antiaromatic system.

Q63. Solution

Correct Answer: (A)

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

Q64. Solution

Correct Answer: (A)

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 1 \times 10^{3}}$$
$$= 3.97 \times 10^{-10} \text{ meter } = 0.397 \text{ nanometer } \approx 0.40 \text{ nm}$$

Q65. Solution

Correct Answer: (B)

Unreacted silver bromide (AgBr) can be dissolved from photographic emulsions using 'hypo' solution $(Na_2\ S_2O_3)$ (aq.). The reaction can be represented as:
$AgBr(s) + 3Na_2\ S_2O_3(aq) \rightarrow Na_5Ag(S_2O_3)_3(aq) + NaBr(aq)$

Q66. Solution

Correct Answer: (A)

Fructose reduces Tollen's reagent because of enolisation of fructose followed by conversion to aldehyde by base. Explaination Fructose is a ketone, but only aldehydes reduce Tollen's reagent. In aqueous solution, fructose is enolised and then converted into aldehyde in basic medium. Aldehydes are organic compounds with a functional group of the structure (R-CH=O)

Q67. Solution

Correct Answer: (C)

C_2H_2 forms copper acetylide with ammonical Cu_2Cl_2
$Cu_2Cl_2 + 2NH_4OH \rightarrow Cu_2O + H_2O + 2NH_4Cl$
$$Cu_2O + C_2H_2 \rightarrow \underset{\text{(copper acetylide)}}{Cu_2C_2} + H_2O$$

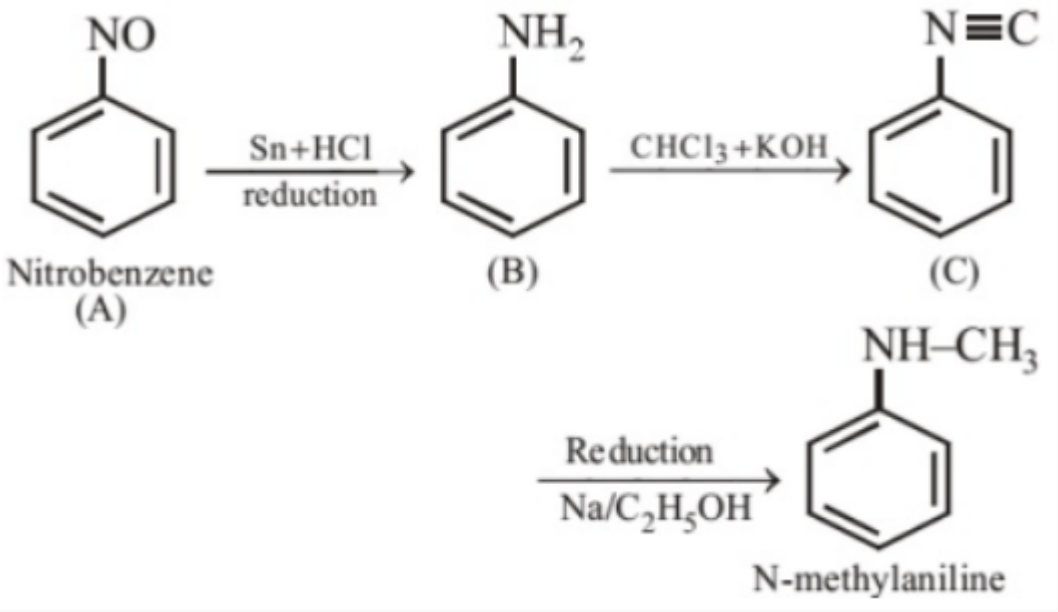
Q68. Solution

Correct Answer: (D)

Vitamin E is an antioxidant present in edible oils.

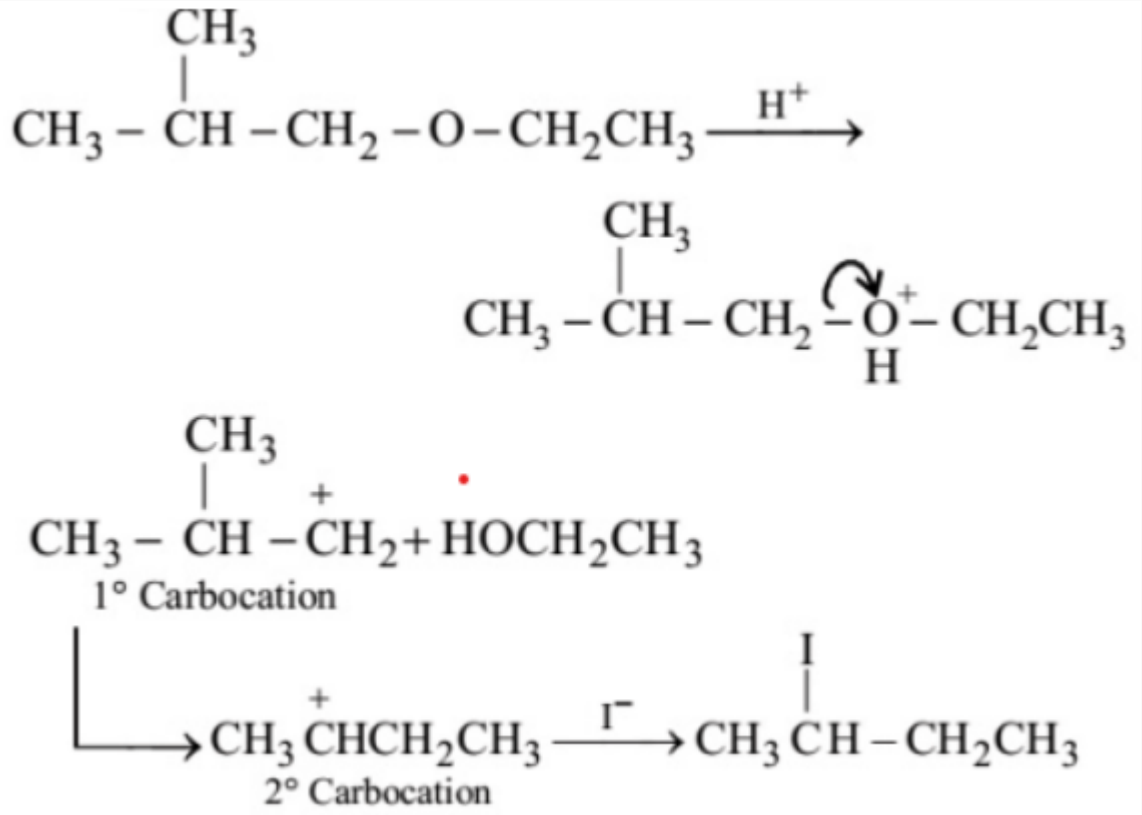
Q69. Solution

Correct Answer: (C)



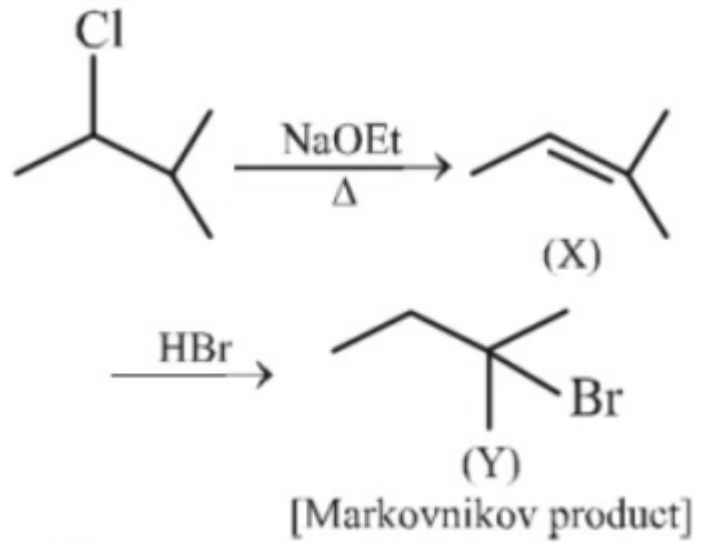
Q70. Solution

Correct Answer: (B)



Q71. Solution

Correct Answer: (C)



Q72. Solution

Correct Answer: (A)

Clean water has BOD value less than 5ppm. Polluted water has BOD value higher than 10ppm.

Q73. Solution

Correct Answer: (B)

Ionic compounds have high melting point. Greater the ionic character, more is melting point. HCl has least ionic character because of maximum electronegativity difference between the two constituent elements, (H and Cl) among $\text{CsF}, \text{HCl}, \text{HF}$ and LiF

$\therefore$ HCl has minimum melting point.

Q74. Solution

Correct Answer: (B)

Under identical conditions, $\frac{r_1}{r_2} = \sqrt{\frac{\text{M}_2}{\text{M}_1}}$

As rate of diffusion is also inversely proportional to time, we will have, $\frac{t_2}{t_1} = \sqrt{\frac{\text{M}_2}{\text{M}_1}}$

(A) Thus, For $\text{He}, t_2 = \sqrt{\frac{4}{2}}(5s) = 5\sqrt{2}s \neq 10s$;

(B) For$\text{O}_2, t_2 = \sqrt{\frac{32}{2}}(5s) = 20s$

(C) For $\text{CO}, t_2 = \sqrt{\frac{28}{2}}(5s) \neq 25s$;

(D) For $\text{CO}_2, t_2 = \sqrt{\frac{44}{2}}(5s) \neq 55s$

#

Q75. Solution
Correct Answer: (B)

$$q = -w_{rev} = -\left(-2.303nRT \log \frac{P_1}{P_2}\right)$$
$$= 2.303 \times \frac{40}{4} \times 2 \times 300 \log \frac{1}{10} = -13.82\text{kcal}$$

Q76. Solution
Correct Answer: (B)

	PCl_5	$\rightarrow$	PCl_3	+	Cl_2
Moles at equilibrium	$\frac{1}{2}$		$\frac{1}{2}$		$\frac{1}{2}$
Mole fraction at equilibrium	$\frac{1}{3}$		$\frac{1}{3}$		$\frac{1}{3}$
Partial pressure at equilibrium	$\frac{P}{3}$		$\frac{P}{3}$		$\frac{P}{3}$

$$K_p = \frac{\frac{P}{3} \times \frac{P}{3}}{\frac{P}{3}} = \frac{P}{3}$$

Q77. Solution
Correct Answer: (C)

The balanced equation :
$2NH_3 + OCl^- \rightarrow N_2H_4 + Cl^- + H_2O$

Q78. Solution
Correct Answer: (D)

Li and Mg do not form solid bicarbonate, but react with N_2 to give nitrides.
$6Li + N_2 \xrightarrow{\Delta} 2Li_3\,N$
$3Mg + N_2 \xrightarrow{\Delta} Mg_3\,N_2$

Q79. Solution
Correct Answer: (A)

$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$
$B_2H_6 + 6H_2O \rightarrow 2H_3BO_3 + 6H_2$

Q80. Solution

Correct Answer: (B)

$\Delta^{\circ}_{Na^+} + \Delta^{\circ}_{OH^-} = 248 \times 10^{-4} Sm^2\ mol^{-1}\ \Delta^{\circ}_{Na^+} + \Delta^{\circ}_{Cl^-} = 126 \times 10^{-4} Sm^2\ mol^{-1}$
$\Delta^{\circ}_{Ba}{}^{2+} + \Delta^{\circ}_{2Cl^-} = 280 \times 10^{-4} Sm^2\ mol^{-1}$ NoW, $\Delta^{\circ}_{Ba(OH)_2} = \Delta^{\circ}_{BaCl_2} + 2\Delta^{\circ\circ}{}_{NaOH} - 2\Delta^{\circ}_{NaC1}$
$\Delta^{\circ}_{Ba(OH)_2} = 280 \times 10^{-4} + 2 \times 248 \times 10^{-4} - 2 \times 126 \times 10^{-4}\ \Delta^{\circ}_{Ba(OH)_2} = 524 \times 10^{-4} Sm^2\ mol^{-1}$

Q81. Solution

Correct Answer: (D)

The correct sequence 'is 'If Socrates was innocent at the age of seventy it may be imagined now innocent St. John was at the age of seventeen'.
Therefore correct combination is d.

Q82. Solution

Correct Answer: (C)

The correct sequence is 'It was all very wonderful and glamorous here in the old places that seemed so ordinary'.
Therefore correct combination is c.

Q83. Solution

Correct Answer: (D)

The correct sequence is If you feed a dog or tame a bear by hand, they get their teeth into the meet and tear and pull at it until they bite a piece of or until they succeed in getting it all out of our hand'.
Therefore correct combination is (d).

Q84. Solution

Correct Answer: (B)

To express general truths we use simple present tense, therefore, (b) is the correct choice.

Q85. Solution

Correct Answer: (B)

The simple past is used to indicate an action completed in the past. Therefore option(b) is correct.

Q86. Solution

Correct Answer: (A)

Simple present tense is also used, instead to the simple future tense, therefore option (a) is correct.

Q87. Solution
Correct Answer: (D)

'Amenities' means 'pleasing acts' and 'courtesies' means 'polite behaviour'. Therefore correct synonym is (d).

Q88. Solution
Correct Answer: (C)

'deflect' means 'turn aside' and 'pervert' means 'twist'. Therefore correct synonym is (c).

Q89. Solution
Correct Answer: (C)

'exorbitant' means 'beyond the limit' and 'excessive' means 'extreme'. Therefore correct synonym is (c).

Q90. Solution
Correct Answer: (D)

'at' is used for things at rest.

Q91. Solution
Correct Answer: (C)

'towards' is used to show the sense of direction.

Q92. Solution
Correct Answer: (A)

'at' is used for things at rest.

Q93. Solution
Correct Answer: (B)

According to the passage, our world has a heritage of culture means, we have an inherited cultural tradition. Therefore option (b) is correct.

Q94. Solution
Correct Answer: (C)

According to the passage common people may be described as the less active and important people. Therefore option (c) is close to correct answer.

Q95. Solution

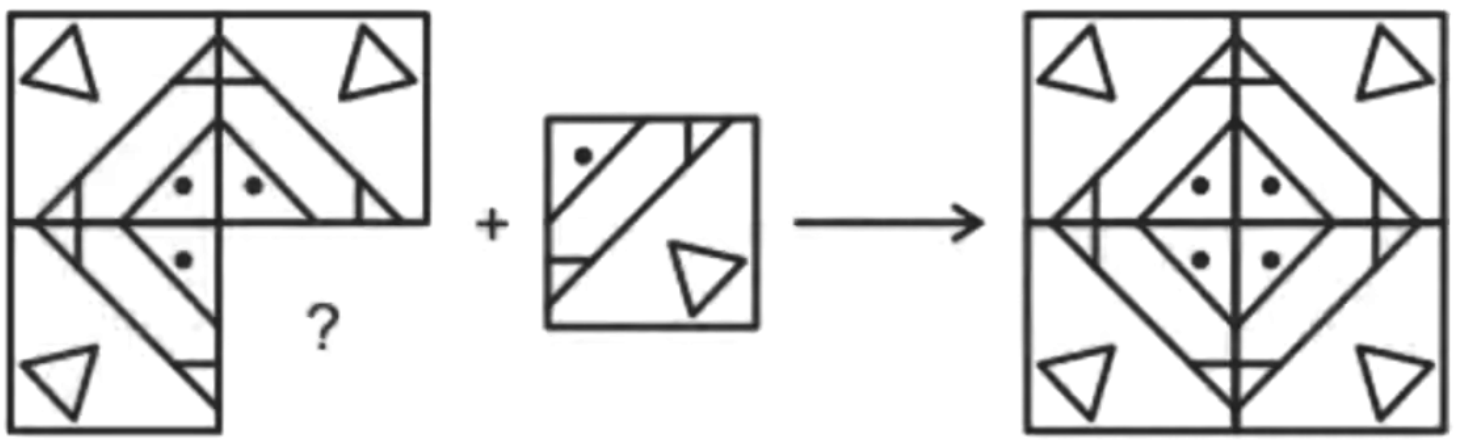
Correct Answer: (C)

According to the passage the persons who are not aware of history are the men ignorant of the past. Therefore option (c) is correct.

Q96. Solution

Correct Answer: (D)

The figure will complete the pattern in the question figure is,

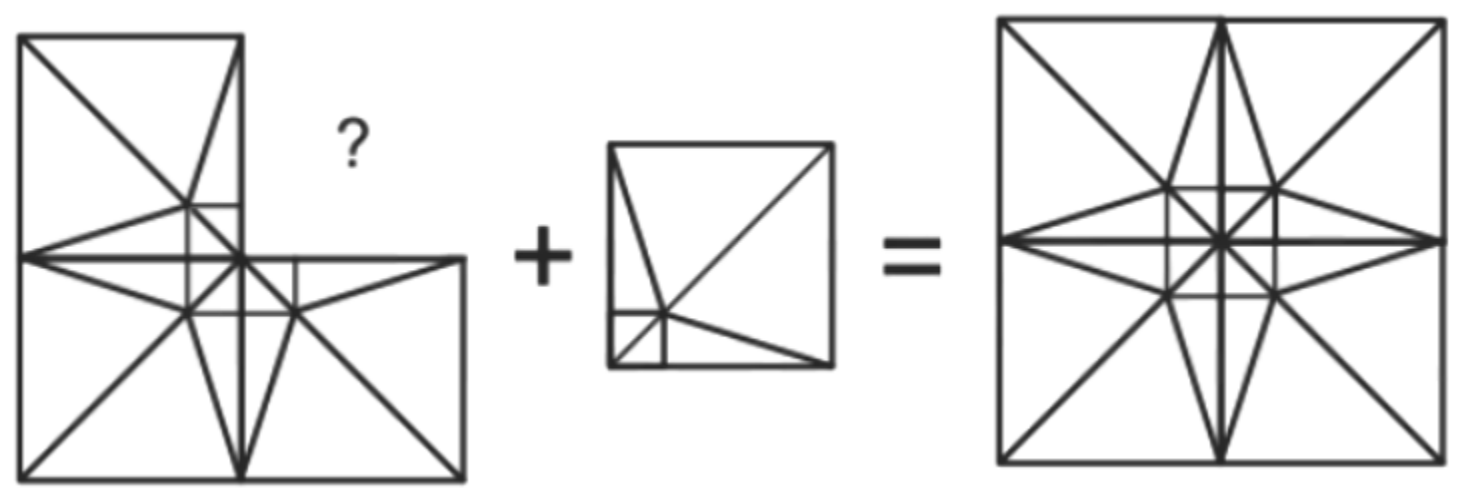


Hence, option 4 is the correct answer.

Q97. Solution

Correct Answer: (D)

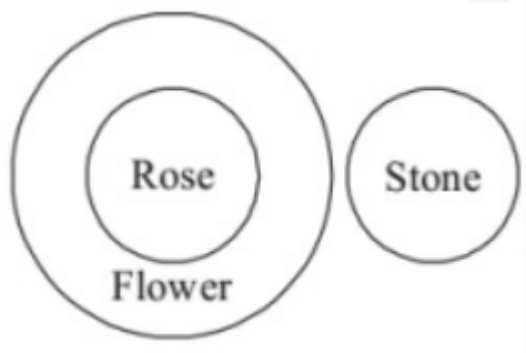
When the image is placed in the blank space, we get:



Hence, the correct answer is "Option 4".

Q98. Solution

Correct Answer: (A)



All roses are flower. Stone is neither Flower nor Rose.1

Q99. Solution

Correct Answer: (A)

According to the given figure: The figure Appear when it is opened as follows:

Hence, option (2) is the correct answer.

Q100. Solution

Correct Answer: (D)

$P \Rightarrow 2$

$N \Rightarrow 4$

$L \Rightarrow 6$

$J \Rightarrow 8$

$P \underset{+1}{+} Q \Rightarrow Q = 2 - 1 = 1$

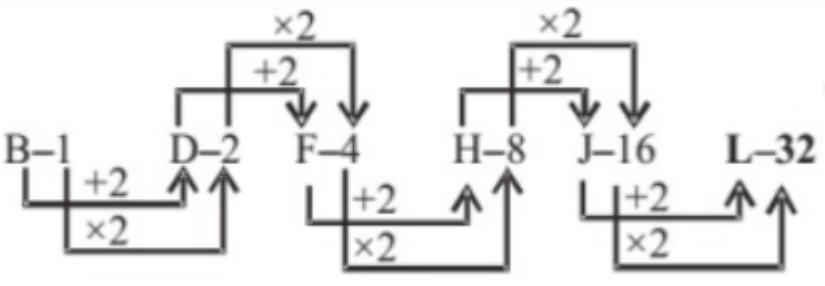
$N = O \Rightarrow O = 4 - 1 = 3$

$J_{+1} K \Rightarrow K = 8 - 1 = 7$

$L = 6$ (Given)

Q101. Solution

Correct Answer: (B)



Q102. Solution

Correct Answer: (C)

Arrange all the vowels and then consonants from left. IOUFHLRS Now all the vowels are replaced with previous letters and all the consonants are replaced by next letter. HNTGIMST Now third letter ...will be M

Q103. Solution

Correct Answer: (D)

The female members in the family are mother, wives of 3 married son's unmarried daughter and 2 daughters of each of the two sons. $(1+3+1+2\times 2)=9$

Q104. Solution

Correct Answer: (B)

Q105. Solution

Correct Answer: (D)

$\left(\frac{3}{2}-1\right)=\frac{1}{2}\Rightarrow 1^{\text{st}}$ row
$\left(\frac{8}{3}-2\right)=\frac{2}{3}\Rightarrow 2^{\text{nd}}$ row
$\left(\frac{19}{5}-3\right)=\frac{4}{5}\Rightarrow 3^{\text{rd}}$ row

Q106. Solution

Correct Answer: (C)

$a=25, d=22-25=-3.$

Let n be the no. of terms

$\text{Sum } = 116;$

$\text{Sum } = \frac{n}{2}[2a+(n-1)d]$

$116=\frac{n}{2}[50+(n-1)(-3)]$

or $232=n[50-3n+3]=n[53-3n]=-3n^2+53n$

$\Rightarrow 3n^2-53+232=0$

$\Rightarrow (n-8)(3n-29)=0$

$\Rightarrow n=8$ or $n=\frac{29}{3}$,

$n \neq \frac{29}{3}$

$\therefore n=8$

$\therefore$ Now, $\mathrm{T}_8=a+(8-1)d=25+7\times(-3)=25-21=4$

$\therefore$ Last term $=4$

Q107. Solution

Correct Answer: (C)

Given parallel lines are

$3x-4y+7=0$ and $3x-4y+5=0$

Required distance $=\frac{|7-5|}{\sqrt{(3)^2+(-4)^2}}=\frac{2}{5} \Rightarrow a=2, b=5$

Q108. Solution

Correct Answer: (A)

We have, $\lim_{x\to 0}\frac{a^{\sin x}-1}{\sin x}=\lim_{y\to 0}\frac{a^y-1}{y}=\log \mathrm{a}$, where $\mathrm{y}=\sin \mathrm{x}$

$\because \mathrm{x}\to 0 \Rightarrow \mathrm{y}=\sin \mathrm{x}\to 0$

Q109. Solution

Correct Answer: (B)

$\int \frac{\sin x}{\sin(x-\alpha)}\mathrm{dx}=\int \frac{\sin(x-\alpha+\alpha}{\sin(x-\alpha)}\mathrm{dx}$

$=\int \frac{\sin(x-\alpha)\cos\alpha+\cos(x-\alpha)\sin\alpha}{\sin(x-\alpha)}\mathrm{dx}$

$=\int\{\cos\alpha+\sin\alpha\cot(x-\alpha)\}\mathrm{dx}$

$=(\cos\alpha)x+(\sin\alpha)\log\sin(x-\alpha)+\mathrm{C}$

$\therefore \mathrm{A}=\cos\alpha, \mathrm{B}=\sin\alpha$

Q110. Solution

Correct Answer: (D)

f(x) is defined for $x^{12}-x^{9}+x^{4}-x+1>0$

$\Rightarrow x^{4}\left(x^{8}+1\right)-x\left(x^{8}+1\right)+1>0$

$\Rightarrow \left(x^{8}+1\right)x\left(x^{3}-1\right)+1>0$

If $x \geq 1$ or $x \leq -1$, then the above expres - sion is positive.

If $-1<x \leq 0$, the above inequality still holds.

If $0<x<1$, then

$x^{12}-x\left(x^{8}+\right)+\left(x^{4}+1\right)>0$

$\because\left[x^{4}+1>x^{8}+1 \text{ and so } x^{4}+1>x\left(x^{8}+1\right)\right]$

The domain of $f=(-\infty, \infty)$

Q111. Solution

Correct Answer: (C)

Let a be the major axis and b, the minor axis of the ellipse, then 3 minor axis = major axis.

$\Rightarrow 3b=a$

Eccentricity is given by:

$b^{2}=a^{2}\left(1-e^{2}\right)$

$\Rightarrow b^{2}=9b^{2}\left(1-e^{2}\right)$

$\Rightarrow \frac{1}{9}=\left(1-e^{2}\right)$

$\Rightarrow e^{2}=\frac{8}{9}$

$\Rightarrow e=\frac{2\sqrt{2}}{3}$

Q112. Solution

Correct Answer: (A)

Sum of odd numbers between 1 and 1000 , which is divisible by $3=3+9+15+21+27+\ldots+999=\mathrm{S}$ (let)

$\therefore$ Let n be the number of terms in series and a is first term.

$\therefore l=a+(n-1)d$,

where l is last term and d is is common difference.

$999=3+(n-1)\times 6$

$n-1=\frac{999-3}{6}=\frac{996}{6}$

$\Rightarrow n-1=166$

$\Rightarrow n=167$

$\therefore \mathrm{S}=\frac{n}{2}[2a+(n-1)d]$

$=\frac{167}{2}[2\times 3+(167-1)\times 6]$

$=\frac{167}{2}[1002]=167\times 501=83667$

Q113. Solution
Correct Answer: (C)

The numbers between 999 and 10000 are all 4-digit numbers. The number of 4 - digit numbers formed by digits $0, 2, 3, 6, 7, 8$ is ${}^{6}\mathrm{P}_4 = 360$. But here those numbers are also involved which begin from 0 . So, we take those numbers as three-digit numbers.
Taking initial digit 0 , the number of ways to fill remaining 3 places from five digits $2, 3, 6, 7, 8$ are ${}^{5}\mathrm{P}_3 = 60$
So, the required numbers $= 360 - 60 = 300$.

Q114. Solution
Correct Answer: (A)

Let $\mathrm{P}(0, 7, 10), \mathrm{Q}(-1, 6, 6)$ and $\mathrm{R}(-4, 9, 6)$ be the vertices of a triangle Here, $\mathrm{PQ} = \sqrt{1+1+16} = 3\sqrt{2}$
$\mathrm{QR} = \sqrt{9+9+0} = 3\sqrt{2}$
$\mathrm{PR} = \sqrt{16+4+16} = 6$
Now, $\mathrm{PQ}^2 + \mathrm{QR}^2 = (3\sqrt{2})^2 + (3\sqrt{2})^2 = 36 = (\mathrm{PR})^2$
Therefore, $\triangle\mathrm{PQR}$ is a right angled triangle at Q. Also, $\mathrm{OQ} = \mathrm{QR}$. Hence, $\triangle\mathrm{PQR}$ is a right angled isosceles triangle.

Q115. Solution
Correct Answer: (C)

Let α, β be the roots of the equation.
$\therefore \alpha + \beta = a - 2$ and $\alpha\beta = -(a+1)$
Now, $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$
$= (a-2)^2 + 2(a+1) = (a-1)^2 + 5$
$\therefore \alpha^2 + \beta^2$ will be minimum if $(a-1)^2 = 0$,
i.e. $a = 1$.

Q116. Solution
Correct Answer: (B)

Let
$\mathrm{I} = \int \cos(\log \mathrm{x}) \cdot 1 \, \mathrm{dx} \ldots\ldots$ (i)
Use integral by parts,

$$I = \cos(\log x) \cdot x - \int [-\sin(\log x)] \cdot \frac{1}{x} \cdot x dx$$

$$= x \cdot \cos(\log x) + \int \sin(\log x) \cdot 1 dx$$

$$= x \cdot \cos(\log x) + \left[\sin(\log x) \cdot x - \int \cos(\log x) \cdot \frac{1}{x} \cdot x dx\right] + C$$

$$= x \cdot (\log x) + \left[x \sin(\log x) - \int \cos(\log x) dx\right] + C$$

$$= x\{\sin(\log x) + \cos(\log x)\} - I + C[from \text{ Eq. (i) }]$$

$$\Rightarrow I = \frac{x}{2}\{\sin(\log x) + \cos(\log x)\} + C$$

Q117. Solution

Correct Answer: (C)

Let feet of perpendicular is

$(2\alpha, 3\alpha+2, 4\alpha+3)$

$\Rightarrow$ Direction ratio of the $\perp$ line is

$2\alpha-3, 3\alpha+3, 4\alpha-8$. and

$\Rightarrow$ Direction ratio of the line are $2, 3, 4$

$\Rightarrow 2(2\alpha-3)+3(3\alpha+3)+4(4\alpha-8)=0$

$\Rightarrow \alpha=1$

$\Rightarrow$ Feet of $\perp$ is $(2,5,7)$

$\Rightarrow$ Length $\perp$ is $\sqrt{1^2+6^2+4^2}=\sqrt{53}$

Q118. Solution

Correct Answer: (A)

According to the condition,

$\frac{\mathrm{dy}}{\mathrm{dx}}=\frac{y}{x}-\cos^2\frac{y}{x}\ldots$ (i)

This is a homogeneous differential equation Substituting $y=vx$, we get

$v+x\frac{\mathrm{dv}}{\mathrm{dx}}=v-\cos^2 v$

$\Rightarrow x\frac{\mathrm{dv}}{\mathrm{dx}}=-\cos^2 v$

$\Rightarrow \int \sec^2 \mathrm{vdv}=-\int \frac{\mathrm{dx}}{\mathrm{x}}$

$\Rightarrow \tan v=-\log x+\mathrm{C}$

$\Rightarrow \tan\frac{y}{x}+\log x=\mathrm{C}$

Substituting $\mathrm{x}=1, y=\frac{\pi}{4}$, we get $\mathrm{C}=1$

Thus, we get

$\tan\left(\frac{y}{x}\right)+\log x=1$

which is the required solution.

Q119. Solution

Correct Answer: (B)

$\frac{1-\cos\alpha+\sin\alpha}{1+\sin\alpha}$

$=\frac{1-\cos\alpha+\sin\alpha}{1+\sin\alpha}\cdot\frac{1+\cos\alpha+\sin\alpha}{1+\cos\alpha+\sin\alpha}$

$=\frac{(1+\sin\alpha)^2-\cos^2\alpha}{(1+\sin\alpha)(1+\cos\alpha+\sin\alpha)}$

$=\frac{(1+\sin^2\alpha+2\sin\alpha)-(1-\sin^2\alpha)}{(1+\sin\alpha)(1+\cos\alpha+\sin\alpha)}$

$=\frac{2\sin\alpha}{1+\cos\alpha+\sin\alpha}=y$

Q120. Solution

Correct Answer: (A)

The $(r+1)$ th term in the expansion of $\left(\sqrt{\frac{x}{3}}+\frac{3}{2x^2}\right)^{10}$ is given by

$\mathrm{T}_{r+1}={}^{10}\mathrm{C}_r\left(\sqrt{\frac{x}{3}}\right)^{10-r}\left(\frac{3}{2x^2}\right)^r$

$={}^{10}\mathrm{C}_r\frac{x^{5-\left(\frac{r}{2}\right)}}{3^{5-\left(\frac{r}{2}\right)}}\cdot\frac{3^r}{2^rx^{2r}}$

$={}^{10}\mathrm{C}_r\frac{3^{\left(\frac{3r}{2}\right)-5}-5-\left(\frac{5r}{2}\right)}{2^r}x$

$\mathrm{T}_{r+1}={}^{10}\mathrm{C}_r\left(\sqrt{\frac{x}{3}}\right)^{10-r}\left(\frac{3}{2x^2}\right)^r={}^{10}\mathrm{C}_r\frac{x^{5-\left(\frac{r}{2}\right)}}{3}\cdot\frac{3^r}{2^rx^{2r}}={}^{10}\mathrm{C}_r\frac{3^{\left(\frac{3r}{2}\right)-5}-\left(\frac{5}{2}\right)-\left(\frac{5r}{2}\right)}{2^r}x$ For T_{r+1} to be independent of x, we must have $5-\left(\frac{5r}{2}\right)=0$ or $r=2$.

Thus, the 3 rd term is independent of x and is equal to ${}^{10}\mathrm{C}_2\frac{3^{3-5}}{2^2}=\frac{10\times9}{2}\times\frac{3^{-2}}{4}=\frac{5}{4}$

Q121. Solution

Correct Answer: (D)

On differentiating

$2^x\log2+2^y\log2\cdot\frac{\mathrm{dy}}{\mathrm{dx}}$

$=2^x\cdot2^y\frac{\mathrm{dy}}{\mathrm{dx}}\cdot\log2+2^y\cdot2^x\log2$

$\Rightarrow2^x+2^y\frac{\mathrm{dy}}{\mathrm{dx}}=2^{x+y}\frac{\mathrm{dy}}{\mathrm{dx}}+2^{x+y}$

$\Rightarrow\frac{\mathrm{dy}}{\mathrm{dx}}=\frac{2^{x+y}-2^x}{2^y-2^{x+y}}$

$\Rightarrow\frac{\mathrm{dy}}{\mathrm{dx}}=\frac{2^x+2^y-2^x}{2^y-2^x-2^y}=-2^{y-x}$

Q122. Solution

Correct Answer: (C)

Let $x=\frac{1-t^2}{1+t^2}$ and $y=\frac{2t}{1+t^2}$

Put $t=\tan\theta$, we get

$x=\frac{1-\tan^2\theta}{1+\tan^2\theta}$ and $y=\frac{2\tan\theta}{1+\tan^2\theta}$

$\Rightarrow x=\cos2\theta$ and $y=\sin2\theta$

$\therefore\frac{\mathrm{dx}}{\mathrm{d}\theta}=-2\sin2\theta$ and $\frac{\mathrm{dy}}{\mathrm{d}\theta}=2\cos2\theta$

Now, $\frac{\mathrm{dy}}{\mathrm{dx}}=\frac{\mathrm{dy}}{\mathrm{d}\theta}\times\frac{\mathrm{d}\theta}{\mathrm{dx}}=-\frac{\cos2\theta}{\sin2\theta}=-\frac{x}{y}$

Q123. Solution

Correct Answer: (D)

The equation of the given curve is

$y = 4x^3 - 2x^5$

$\frac{\mathrm{dy}}{\mathrm{dx}} = 12x^2 - 10x^4$

Therefore, the slope of the tangent at point (x, y) is $12x^2 - 10x^4$. The equation of the tangent at (x, y) is given by

$\mathrm{Y} - y = \left(12x^2 - 10x^4\right)(\mathrm{X} - x) \ldots$ (i)

When, the tangent passes through the origin $(0, 0)$, then $\mathrm{X} = \mathrm{Y} = 0$ Therefore, eq. (i) reduce to

$-y = \left(12x^2 - 10x^4\right)(-x)$

$\Rightarrow y = 12x^3 - 10x^5$

Also, we have $y = 4x^3 - 2x^5$

$\therefore 12x^3 - 10x^5$

$= 4x^3 - 2x^5$

$\Rightarrow 8x^5 - 8x^3 = 0$

$\Rightarrow x^5 - x^3 = 0$

$\Rightarrow x^3\left(x^2 - 1\right) = 0$

$\Rightarrow x = 0, \pm 1$

When, $x = 0$

$y = 4(0)^3 - 2(0)^5 = 0$

When, $x = 1$,

$y = 4(1)^3 - 2(1)^5 = 2$

When, $x = -1$,

$y = 4(-1)^3 - 2(-1)^5 = -2$

Hence, the require points are $(0, 0), (1, 2)$ and $(-1, -2)$.

Q124. Solution

Correct Answer: (C)

Let r be the radius of the sphere and Δr be the error in measuring the radius. Then, $r = 9$ cm and

$\Delta r = 0.03$ cm

Let V be the volume of the sphere. Then,

$\mathrm{V} = \frac{4}{3}\pi r^3$

$\Rightarrow \frac{\mathrm{dV}}{\mathrm{dr}} = 4\pi r^2$

$\Rightarrow \left(\frac{\mathrm{dV}}{\mathrm{dr}}\right)_{r=9} = 4\pi \times 9^2 = 324\pi$

Let $\Delta \mathrm{V}$ be the error in V due to error $\Delta \mathrm{r}$ in r.

Then,

$\Delta \mathrm{V} = \frac{\mathrm{dV}}{\mathrm{dr}} \Delta r$

$\Rightarrow \Delta \mathrm{V} = 324\pi \times 0.03 = 9.72\pi \mathrm{cm}^3$

Q125. Solution

Correct Answer: (C)

Let $z = x + iy$, then its conjugate $\bar{z} = x - iy$

Given that $z^2 = (\bar{z})^2$

$\Rightarrow x^2 - y^2 + 2ixy = x^2 - y^2 - 2ixy$

$\Rightarrow 4ixy = 0$

Q126. Solution

Correct Answer: (B)

If $\mathrm{f}(x)$ is an even function then

$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$

Here $\mathrm{f}(x) = x^8 - x^4 + x^2 + 1$ is an even function, therefore $a = 4$.

Q127. Solution

Correct Answer: (D)

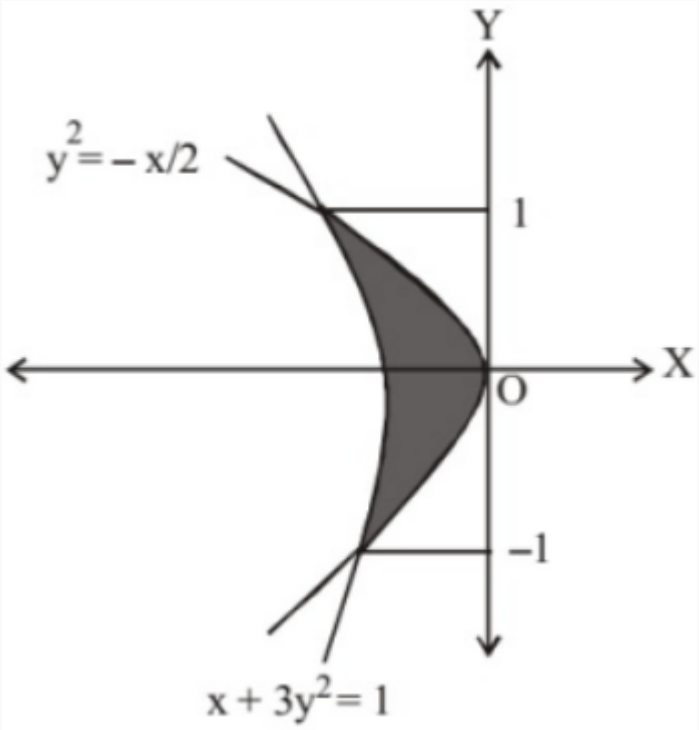
We have, $x + 2y^2 = 0$

$\Rightarrow y^2 = -\frac{x}{2} \ldots$ (i)

a parabola with vertex $(0, 0)$ and $x + 3y^2 = 1$

$\Rightarrow y^2 = \frac{1-x}{3} = -\left(\frac{x-1}{3}\right) \ldots$ (ii), a parabola with vertex $(1, 0)$

Solving (i) and (ii), we get $y = \pm 1$



$\mathrm{A} = \int_{-1}^{1} \left(\left(1 - 3y^2\right) - \left(-2y^2\right)\right) dy$

$= 2\int_{0}^{1} \left(1 - y^2\right) dy = 2\left[y - \frac{y^3}{3}\right]_0^1$

$= \frac{4}{3}$ sq.units

Q128. Solution

Correct Answer: (A)

We have $\frac{d}{dx}\left(\tan^{-1}\sqrt{\frac{1+\cos\frac{x}{2}}{1-\cos\frac{x}{2}}}\right)$

$= \frac{d}{dx}\left[\tan^{-1}\sqrt{\frac{1+\left(2\cos^2\frac{x}{4}-1\right)}{1-\left(1-2\sin^2\frac{x}{4}\right)}}\right]$

$= \frac{d}{dx}\left(\tan^{-1}\sqrt{\frac{2\cos^2\frac{x}{4}}{2\sin^2\frac{x}{4}}}\right)$

$= \frac{d}{dx}\left(\tan^{-1}\sqrt{\cot^2\frac{x}{4}}\right) = \frac{d}{dx}\left(\tan^{-1}\cot\frac{x}{4}\right)$

$= \frac{d}{dx}\left(\tan^{-1}\tan\left(\frac{\pi}{2}-\frac{x}{4}\right)\right) = \frac{d}{dx}\left(\frac{\pi}{2}-\frac{x}{4}\right) = 0-\frac{1}{4} = -\frac{1}{4}$

Q129. Solution

Correct Answer: (A)

$\because \mathrm{f}(x) = \left(\frac{e^{2x}-1}{e^{2x}+1}\right)$

$\therefore \mathrm{f}(-x) = \frac{e^{-2x}-1}{e^{-2x}+1} = \frac{1-e^{2x}}{1+e^{2x}}$

$\Rightarrow \mathrm{f}(-x) = \frac{-\left(e^{2x}-1\right)}{e^{2x}+1} = -\mathrm{f}(x)$

$\therefore \mathrm{f}(x)$ is an odd function.

Again, $\mathrm{f}'(x) = \frac{4e^{2x}}{\left(1+e^{2x}\right)^2} > 0, \forall x \in R \Rightarrow \mathrm{f}(x)$ is an increasing function.

Q130. Solution

Correct Answer: (A)

We have, $\int_0^{\pi}\frac{1}{5+4\cos x}dx$

$= \int_0^{\pi}\frac{1}{5+4\left(\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)}dx$

$= \int_0^{\pi}\frac{1+\tan^2\frac{x}{2}}{5\left(1+\tan^2\frac{x}{2}\right)+4\left(1-\tan^2\frac{x}{2}\right)}dx$

$= \int_0^{\pi}\frac{1+\tan^2\frac{x}{2}}{9+\tan^2\frac{x}{2}}\mathrm{dx} = \int_0^{\pi}\frac{\sec^2\frac{x}{2}}{9+\tan^2\frac{x}{2}}\mathrm{dx}$

Let $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

Also, $x = 0 \Rightarrow t = 0$ and $x = \pi \Rightarrow t = \infty$

$\therefore \mathrm{I} = \int_0^{\infty}\frac{\mathrm{dt}}{9+t^2}$

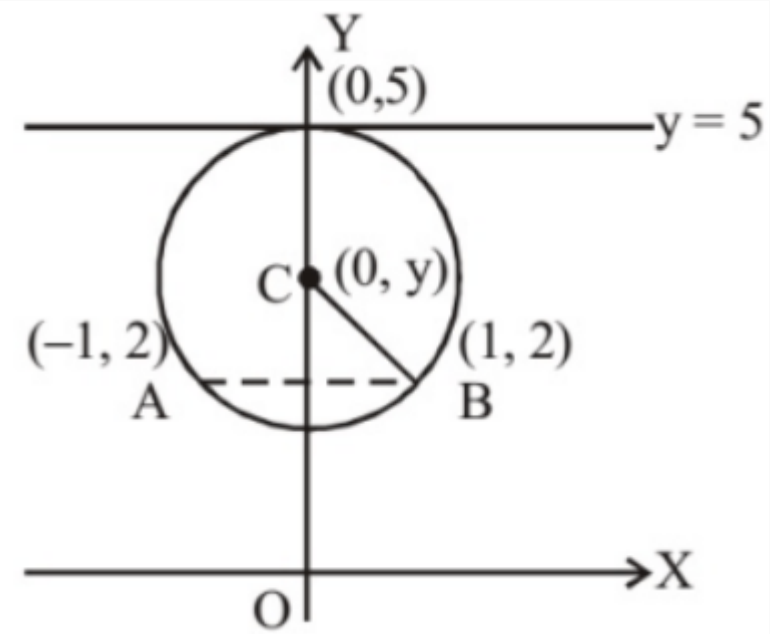
$\therefore \mathrm{I} = 2\int_0^{\infty}\frac{\mathrm{dt}}{9+t^2}$

$\therefore \mathrm{I} = \frac{2}{3}\left[\tan^{-1}\frac{t}{3}\right]_0^{\infty} = \frac{2}{3}\left[\tan^{-1}\infty - \tan^{-1}0\right]$

$= \frac{2}{3}\left(\frac{\pi}{2}-0\right) = \frac{\pi}{3}$

Q131. Solution

Correct Answer: (A)



The centre of the circle is on the perpendicular bisector of the line joining $(-1, 2)$ and $(1, 2)$, which is the y-axis.
The ordinate of the centre is given by
$(5-y)^2 = 1 + (y-2)^2$
$\Rightarrow y = \frac{10}{3}$
Hence, eq. of the circle is:
$x^2 + \left(y - \frac{10}{3}\right)^2 = \left(\frac{5}{3}\right)^2$
$\Rightarrow 9x^2 + 9y^2 - 60y + 75 = 0$

Q132. Solution

Correct Answer: (A)

Expression $= (1-x)^5 \cdot (1+x)^4 \left(1+x^2\right)^4$
$= (1-x)\left(1-x^2\right)^4 \left(1+x^2\right)^4$
$= (1-x)\left(1-x^4\right)^4$
$\therefore$ Coefficient of $x^{13} = -{}^4\mathrm{C}_3(-1)^3 = 4$

Q133. Solution

Correct Answer: (B)

The given differential equation can be written as $\frac{dx}{dy}+\frac{x}{1+y^2}=\frac{\tan^{-1}y}{1+y^2}\ldots$ (i)

Now, eq. (i) is a linear differential equation of the form $\frac{dx}{dy}+\mathrm{P}_1x=\mathrm{Q}_1$

where $\mathrm{P}_1=\frac{1}{1+y^2}$ and $\mathrm{Q}_1=\frac{\tan^{-1}y}{1+y^2}$

Therefore, I.F $=e^{\int\frac{1}{1+y^2}dy}=e^{\tan^{-1}y}$

Thus, the solution of the given differential equation is given by $xe^{\tan^{-1}y}=\int\left(\frac{\tan^{-1}y}{1+y^2}\right)e^{\tan^{-1}y}\mathrm{dy}+\mathrm{C}\ldots$ (ii)

Let $\mathrm{I}=\int\left(\frac{\tan^{-1}y}{1+y^2}\right)e^{\tan^{-1}y}\mathrm{dy}$

On substituting $\tan^{-1}y=t$, so that

$\left(\frac{1}{1+y^2}\right)\mathrm{dy}=\mathrm{dt}$, we get

$I=\int te^tdt=te^t-\int 1\cdot e^tdt=te^t-e^t=e^t(t-1)$

or $\mathrm{I}=e^{\tan^{-1}y}\left(\tan^{-1}y-1\right)$

On substituting the value of 1 in equation (i1), we get

$x\cdot e^{\tan^{-1}y}=\cdot e^{\tan^{-1}y}\left(\tan^{-1}y-1\right)+\mathrm{C}$

or $x=\left(\tan^{-1}y-1\right)+\mathrm{C}e^{\tan^{-1}y}$

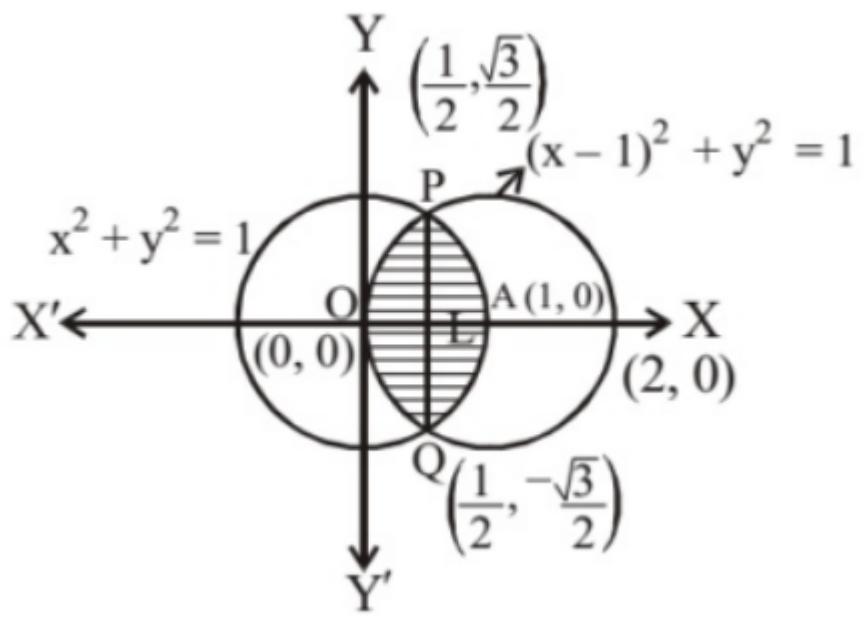
which is the general solution of the given differntial equation.

Q134. Solution

Correct Answer: (A)

Given circles are $x^2+y^2=1 \ldots$ (i) and $(x-1)^2+y^2=1 \ldots$ (ii)

Centre of (i) is $\mathrm{O}(0,0)$ and radius $=1$



Both these circle are symmetrical about x-axis solving (i) and (ii), we get, $-2x+1=0$

$\Rightarrow x=\frac{1}{2}$

then $y^2=1-\left(\frac{1}{2}\right)^2=34$

$\Rightarrow y=\frac{\sqrt{3}}{2}$

$\therefore$ The points of intersection are

$\mathrm{P}\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$ and $\mathrm{Q}\left(\frac{1}{2},-\frac{\sqrt{3}}{2}\right)$

It is clear from the figure that the shaded portion in region whose area is required.

$\therefore$ Required area = area OQAPO

$=2\times$ area of the region OLAP

$=2\times$ (area of the region $OLPO+$ area of $LAPL$)

$=2\left[\int_0^{\frac{1}{2}}\sqrt{1-(x-1)^2}dx+\int_{\frac{1}{2}}^1\sqrt{1-x^2}\mathrm{dx}\right]$

$=2\left[\frac{(x-1)\sqrt{1-(x-1)^2}}{2}+\frac{1}{2}\sin^{-1}(x-1)\right]_0^{\frac{1}{2}}+2\left[\frac{x\sqrt{1-x^2}}{2}+\frac{1}{2}\sin^{-1}x\right]_{\frac{1}{2}}^1$

$=-\frac{1}{2}\cdot\frac{\sqrt{3}}{2}+\sin^{-1}\left(\frac{-1}{2}\right)-\sin^{-1}(-1)+0+\sin^{-1}(1)-\left(\frac{1}{2}\cdot\frac{\sqrt{3}}{2}+\sin^{-1}\left(\frac{1}{2}\right)\right)$

$=\left(\frac{2\pi}{3}-\frac{\sqrt{3}}{2}\right)$ sq. units.

Q135. Solution

Correct Answer: (D)

Since, sum $=4$ and second term $=\frac{3}{4}\Rightarrow\frac{a}{1-r}=4$, and $ar=\frac{3}{4}\Rightarrow\frac{a}{1-\frac{3}{4a}}=4\Rightarrow(a-1)(a-3)=0\Rightarrow a=1$ or $a=3$

$\Rightarrow\frac{a}{1-\frac{3}{4a}}=4$

$\Rightarrow(a-1)(a-3)=0$

$\Rightarrow a=1$ or $a=3$

Q136. Solution

Correct Answer: (B)

$\int \frac{e^x(1+x)}{\cos^2(e^x x)}\mathrm{dx}$

Let $xe^x = t$

$\Rightarrow (xe^x + e^x) = \frac{\mathrm{dt}}{\mathrm{dx}}$

$\Rightarrow dx = \frac{\mathrm{dt}}{e^x(x+1)}$

$\therefore \int \frac{e^x(1+x)}{\cos^2(e^x x)}\mathrm{dx} = \int \frac{e^x(1+x)}{\cos^2 t} \times \frac{dt}{e^x(1+x)} = \int \frac{1}{\cos^2 t}\mathrm{dt} = \int \sec^2 t\mathrm{dt} = \tan t + \mathrm{C} = \tan\left(xe^x\right) + \mathrm{C}$

Q137. Solution

Correct Answer: (D)

$\mathrm{B} + \mathrm{C} = \frac{\pi}{2} - \mathrm{A} \Rightarrow \tan(\mathrm{B} + \mathrm{C}) = \cot \mathrm{A}$

$\Rightarrow \frac{\tan \mathrm{B} + \tan \mathrm{C}}{1 - \tan \mathrm{B} \tan \mathrm{C}} = \cot \mathrm{A}$

$\Rightarrow \tan \mathrm{A} \tan \mathrm{B} + \tan \mathrm{B} \tan \mathrm{C} + \tan \mathrm{C} \tan \mathrm{A} = 1$

$\Rightarrow \tan \mathrm{A} \tan \mathrm{B} \tan \mathrm{C}(\cot \mathrm{C} + \cot \mathrm{A} + \cot \mathrm{B}) = 1$

$\Rightarrow \cot \mathrm{A} + \cot \mathrm{B} + \cot \mathrm{C} = \cot \mathrm{A} \cot \mathrm{B} \cot \mathrm{C}$

Again $\cos 2\,\mathrm{A} + \cos 2\,\mathrm{B} + \cos 2\mathrm{C}$

$= 2\cos(\mathrm{A} + \mathrm{B})\cos(\mathrm{A} - \mathrm{B}) + \cos 2\mathrm{C}$

$= 2\cos\left(\frac{\pi}{2} - C\right)\cos(A - B) + 1 - 2\sin^2 C$

$= 2\sin C[\cos(A - B) - \sin C] + 1$

$= 2\sin \mathrm{C}\left[\cos(\mathrm{A} - \mathrm{B}) - \sin\frac{\pi}{2} - (\mathrm{A} + \mathrm{B})\right] + 1$

$= 2\sin C[\cos(A - B) - \cos(A + B)] + 1$

$= 4\sin A \sin B \sin C + 1$

Q138. Solution

Correct Answer: (B)

Consider $\lim_{x\to 0} \frac{2\sin^2 3x}{x^2}$

$= 2 \cdot \lim_{x\to 0}\left[\frac{\sin 3x}{x}\right]^2 = 2 \cdot \lim_{x\to 0}\left[3\frac{\sin 3x}{3x}\right]^2$

$= 2.9\lim_{x\to 0}\left(\frac{\sin 3x}{3x}\right)^2 = 18 \times 1 = 18$

Q139. Solution

Correct Answer: (C)

Putting $\mathrm{H} = \frac{2ab}{a+b}$, we have

$\frac{1}{\mathrm{H}-a} + \frac{1}{\mathrm{H}-b}$

$= \frac{1}{\left(\frac{2ab}{a+b} - a\right)} + \frac{1}{\left(\frac{2ab}{a+b} - b\right)} = \frac{a+b}{ab-a^2} + \frac{a+b}{ab-b^2}$

$= \left(\frac{a+b}{b-a}\right)\left(\frac{1}{a} - \frac{1}{b}\right) = \left(\frac{a+b}{b-a}\right)\left(\frac{b-a}{ab}\right) = \frac{a+b}{ab} = \frac{1}{a} + \frac{1}{b}$

Q140. Solution

Correct Answer: (B)

The desired probability

$= \frac{\frac{10}{100} \times \frac{70}{100}}{\frac{10}{100} \times \frac{70}{100} + \frac{5}{100} \times \frac{30}{100}} = \frac{14}{17}$

Q141. Solution

Correct Answer: (C)

$y^2 = x(2-x)^2 \Rightarrow y^2 = x^3 - 4x^2 + 4x \ldots$ (i)

$\Rightarrow 2y\frac{\mathrm{dy}}{\mathrm{dx}} = 3x^2 - 8x + 4$

$\Rightarrow \frac{\mathrm{dy}}{\mathrm{dx}} = \frac{3x^2-8x+4}{2y}$

$\Rightarrow \left[\frac{\mathrm{dy}}{\mathrm{dx}}\right]_{\mathrm{P}} = \frac{3-8+4}{2} = -\frac{1}{2}$

$y^2 = x(2-x)^2 \Rightarrow y^2 = x^3 - 4x^2 + 4x \ldots$ (i) $\Rightarrow 2y\frac{\mathrm{dy}}{\mathrm{dx}} = 3x^2 - 8x + 4 \Rightarrow \frac{\mathrm{dy}}{\mathrm{dx}} = \frac{3x^2-8x+4}{2y}$

$\Rightarrow \left[\frac{\mathrm{dy}}{\mathrm{dx}}\right]_{\mathrm{P}} = \frac{3-8+4}{2} = -\frac{1}{2}$ $\therefore$ Equation of tangent at P is: $y - 1 = -\frac{1}{2}(x-1) \Rightarrow x + 2y - 3 = 0$

$\Rightarrow x + 2y - 3 = 0$

Using $y = \frac{3-x}{2}$ in (i), we get: $\left(\frac{3-x}{2}\right)^2$

$= x^3 - 4x^2 + 4x$

$\Rightarrow 4x^3 - 17x^2 + 22x - 9 = 0 \ldots$(ii)

which has two roots 1,1 (Because of (ii) being tangent at $(1,1)$).

Sum of 3 roots $= \frac{17}{4}$

$\therefore$ 3rd root $= \frac{17}{4} - 2 = \frac{9}{4}$

Then, $y = \frac{3-\frac{9}{4}}{2} = \frac{3}{8}$

$\therefore$ Q is $\left(\frac{9}{4}, \frac{3}{8}\right)$

Q142. Solution

Correct Answer: (D)

Given $\alpha = 45^\circ, \beta = 60^\circ, \gamma = ?$

$\because \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\therefore \cos^2\gamma = 1 - \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$

$\Rightarrow$ y $= 60^\circ$ or 120°

Q143. Solution

Correct Answer: (A)

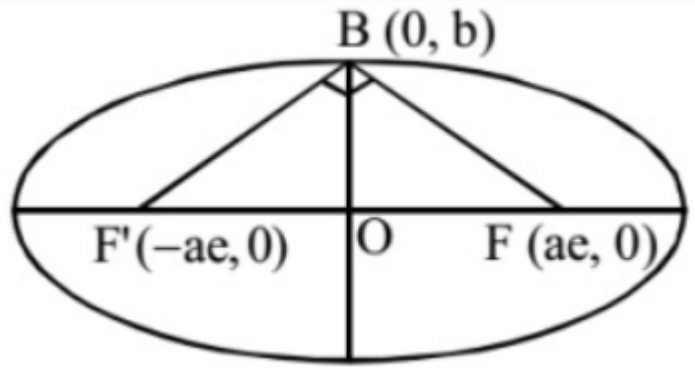
$\because \angle \mathrm{FBF}' = 90^\circ$

$\Rightarrow \mathrm{FB}^2 + \mathrm{F'B}^2 = \mathrm{FF'}^2$

$\therefore \left(\sqrt{a^2e^2+b^2}\right)^2 + \left(\sqrt{a^2e^2+b^2}\right)^2 = (2ae)^2$

$\Rightarrow 2\left(a^2e^2+b^2\right) = 4a^2e^2$

$\Rightarrow e^2 = \frac{b^2}{a^2}$



Also $e^2 = 1 - \frac{b^2}{a^2} = 1 - e^2$

$\Rightarrow 2e^2 = 1$

$\Rightarrow e = \frac{1}{\sqrt{2}}$

Q144. Solution

Correct Answer: (A)

$\cos\left[2\cos^{-1}x + \sin^{-1}x\right]$

$= \cos\left[\cos^{-1}x + \cos^{-1}x + \sin^{-1}x\right]$

$= \cos\left[\cos^{-1}x + \frac{\pi}{2}\right] = -\sin\left[\cos^{-1}x\right]$

$= -\sin\left[\sin^{-1}\sqrt{1-x^2}\right] = -\sqrt{1-x^2}$

$= -\sqrt{1-\left(\frac{1}{5}\right)^2} = -\sqrt{\frac{24}{25}} = -\frac{2\sqrt{6}}{5}$

Q145. Solution

Correct Answer: (A)

Since $\vec{a} = m\vec{b}$ for some scalar m i.e.,

$\vec{a} = m\left(6\hat{i} - 8\hat{j} - \frac{15}{2}\hat{k}\right)$

$\Rightarrow |a| = |m|\sqrt{36 + 64 + \frac{225}{4}}$

$\Rightarrow 50 = \frac{25}{2}|m|$

$\Rightarrow |m| = 4$

$\Rightarrow m = \pm 4$

Since, a makes an acute angle with the positive direction of Z - axis, so its z component must be positive and hence, m must be -4.

$\therefore a = -4\left(6\hat{i} + 8\hat{j} - \frac{15}{2}\hat{k}\right) = -24\hat{i} + 32\hat{j} + 30\hat{k}$

Q146. Solution

Correct Answer: (B)

$$np = 4$$
$$npq = 2$$
$$\Rightarrow q = \frac{1}{2}, p = \frac{1}{2}, n = 8$$
$$\mathrm{P(X = 1)} = {}^{8}\mathrm{C}_1 \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^7 = 8 \cdot \frac{1}{2^8} = \frac{1}{2^5} = \frac{1}{32}$$

Q147. Solution

Correct Answer: (A)

We prove the statement p is true by contrapositive method and by direct method.
Direct method: For any real number x and y,
$x = y$
$\Rightarrow 2x = 2y$
$\Rightarrow 2x + a = 2y + a$ a for some $a \in Z$
Contrapositive method: The contrapositive statement of p is "For any real numbers x, y if $\mathrm{2x + a \neq 2y + a}$, where $\mathrm{a \in Z}$, then $\mathrm{x \neq y}$."
Given, $2x + a \neq 2y + a$
$\Rightarrow 2x \neq 2y$
$\Rightarrow x \neq y$
Hence, the given statement is true.

Q148. Solution

Correct Answer: (A)

$$z_1 = \sqrt{2}\left[\cos\tfrac{\pi}{4} + i\sin\tfrac{\pi}{4}\right] = \sqrt{2}\left[\tfrac{1}{\sqrt{2}} + i\tfrac{1}{\sqrt{2}}\right] = 1 + i$$
$$|z_1| = \sqrt{2}$$
and $z_2 = \sqrt{3}\left[\cos\tfrac{\pi}{3} + i\sin\tfrac{\pi}{3}\right] = \sqrt{3}\left[\tfrac{1}{2} + i\tfrac{\sqrt{3}}{2}\right]$
$$|z_2| = \sqrt{\tfrac{3}{4} + \tfrac{9}{4}} = \sqrt{3}$$
$$|z_1 z_2| = |z_1|\,|z_2| = \sqrt{2} \cdot \sqrt{3} = 6$$

Q149. Solution

Correct Answer: (B)

$$\lim_{x\to 0}\sqrt{\frac{x - \sin x}{x + \sin^2 x}} = \lim_{x\to 0}\sqrt{\frac{1 - \frac{\sin x}{x}}{1 + \frac{\sin^2 x}{x}}}$$
$$= \lim_{x\to 0}\sqrt{\frac{1 - \frac{\sin x}{x}}{1 + \left(\frac{\sin x}{x}\right)\sin x}} = \sqrt{\frac{1-1}{1+1\times 0}} = 0$$

Q150. Solution

Correct Answer: (D)

We have,

$\frac{3x+1}{(x-3)(x-5)} = \frac{-5}{x-3} + \frac{\mathrm{B}}{x-5}$

$3x+1 = -5(x-5) + \mathrm{B}(x-3)$

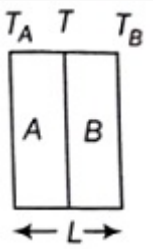
$\text{Put } x = 5$

$3(5)+1 = \mathrm{B}(5-3)$

$16 = 2\,\mathrm{B} \text{ or } \mathrm{B} = 8$

Q151. Solution

Correct Answer: (C)



Let T be the junction temperature

Here, $K_A = 2K_B,\ T - T_B = 50K$

At the steady state $H_A = H_B$

$\Rightarrow \quad \frac{K_A A(T_A - T)}{\frac{L}{2}} = \frac{K_B A\,(T - T_B)}{\frac{L}{2}}$

$\Rightarrow \quad 2K_B\,(T_A - T) = K_B\,(T - T_B)$

$\Rightarrow \quad T_A - T = \frac{T - T_B}{2} = \frac{50}{2} = 25\ K$

Q152. Solution

Correct Answer: (B)

$\theta > \theta_c$

$\sin\theta > \sin\theta_c$

$\sin\theta > \frac{8}{9}.$

Q153. Solution

Correct Answer: (A)

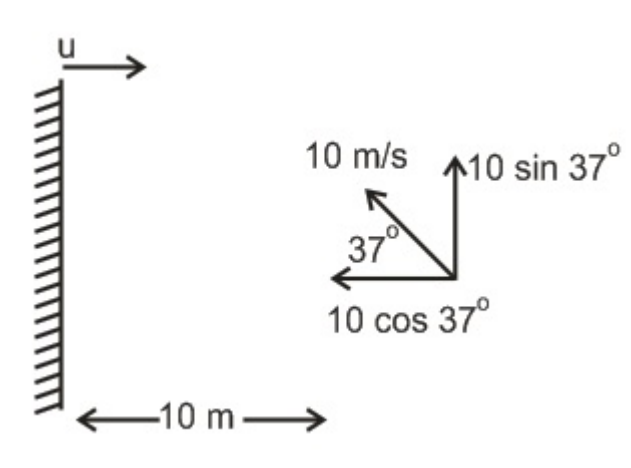
As the vertical motion remains unaffected

$T = \frac{2\,u \sin\,\theta}{g} \quad \Rightarrow \quad T = \frac{2\times10\times3/5}{10}$

$T = \frac{6}{5}s$ ...(1)

from coefficient of restitutation along the line of collision

$e = \frac{\text{relative velocity of separation after collision}}{\text{relative velocity of approach before collision}}$



$1 = \frac{v-u}{8+u}$

v - 2u = 8 ...(2)

by the time (t_1) ball collide with wall, distance travelled by man = 10 - ut_1

$t_1 = \frac{10}{8+u}$ ...(3)

by the time (t_2) ball reach the hand of man $t_2 = \frac{10-u_1}{v}$ ...(4)

Since T = t_1 + t_2 ...(5)

from equation (2), (3), (4), (5)

$u = \frac{13}{3} m/s$

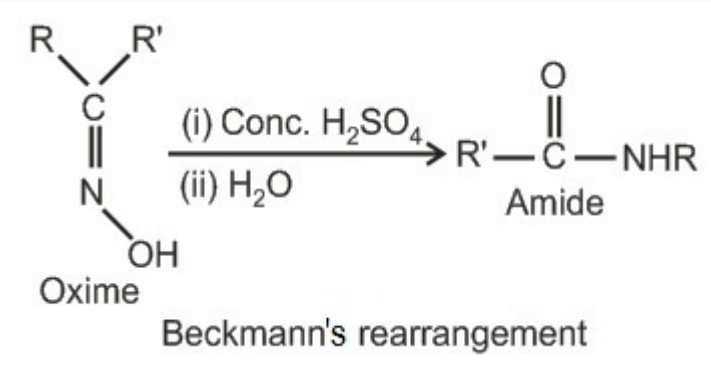
Q154. Solution

Correct Answer: (C)

Magnitude of electric field is constant & the surface is equipotential

Q155. Solution

Correct Answer: (C)



Q156. Solution

Correct Answer: (B)

Beryl is a mineral composed of beryllium aluminum cyclic silicate with the chemical formula $Be_3Al_2(SiO_3)_6$

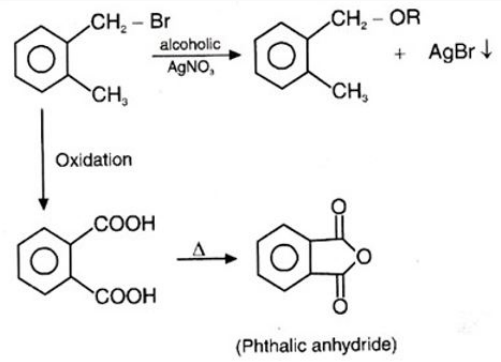
Q157. Solution

Correct Answer: (C)

IUPAC name is dichlorodioxalatochromate (III)

Q158. Solution

Correct Answer: (D)



Q159. Solution

Correct Answer: (C)

$y=\tan^{-1}\left(\frac{\log e-2\log x}{\log e+2\log x}\right)+$

$=\tan^{-1}\left(\dfrac{3+2\log x}{1-6\log x}\right)$

$=\dfrac{\pi}{4}-\tan^{-1}\left(\dfrac{1-2\log x}{1+2\log x}\right)+\tan^{-1}\left(\dfrac{3+2\log x}{1-6\log x}\right)+\tan^{-1}3+\tan^{-1}(2\log x)$

$=\dfrac{\pi}{4}+\tan^{-1}3$

Thus $\frac{dy}{dx}=0 \quad\Rightarrow\quad \frac{d^2y}{dx^2}=0.$

Q160. Solution

Correct Answer: (A)

Using $R_1\leftrightarrow R_3$, write

$$P(x)=-\begin{matrix} x-10 & 3 & 4 \\ 2 & x-10 & 5 \\ 7 & 6 & x-10 \end{matrix}$$

Observe $P(x)$ is of the form $P(x)=-(x-10)^3+a(x-10)+b$ where a,b are some real numbers.

$\therefore$ sum of zeros of $P(x)$

$=-\frac{\text{coefficient of } x^2}{\text{coefficient of } x^3}=30$

Q161. Solution

Correct Answer: (C)

$$f(\theta)=\begin{matrix} -1 & 0 & 1 \\ -1 & 1 & 0 \\ 1+\sin^2\theta & \cos^2\theta & 4\sin 2\theta \end{matrix}$$

$=-\left(\cos^2\theta+1+\sin^2\theta+4\sin 4\theta\right)$

$=-2(1+2\sin 4\theta)$

$(R_1\to R_1-R_3$

$R_2\to R_2-R_3)$

$(C_1\to C_1+C_3)$

$$=\begin{matrix} 0 & 0 & 1 \\ -1 & 1 & 0 \\ 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \end{matrix}$$

$+4\sin 4\theta$

$=-\left(\cos^2\theta+1+\sin^2\theta+4\sin 4\theta\right)$

$=-2(1+2\sin 4\theta)$

which is periodic function with period $\frac{2\pi}{4}=\frac{\pi}{2}$.

Q162. Solution

Correct Answer: (B)

Diameter of $(x-1)^2+y^2=1$ is $2\times 1=2=b$ and of $x^2+(y-2)^2=4$ is $2\times 2=4=a$

So the required equation of the ellipse is

$\frac{x^2}{(4)^2}+\frac{y^2}{(2)^2}=1\Rightarrow x^2+4y^2=16$